
How Visual Sensor Shapes the Format of Spatial Memory in Navigation Agents

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Abstract

Recent lines of work on foundation visual encoders, vision–language models, transformer world models, and embodied LLM agents share an implicit premise: that richer perception translates uniformly into better downstream reasoning. Yet vision–language models with strong surface-level recognition still fail at fine-grained spatial reasoning. The default account is a training-distribution gap; we examine a structural alternative that frontier systems are too architecturally complex to test cleanly. The question has biological precedent: hippocampal spatial coding reorganises with sensor structure across species, from foveated primates to echolocating bats. A recurrent navigation agent serves as a controlled chassis — a single, probeable hidden state, with everything held identical except the visual sensor, varied along a single axis of encoder spatial bandwidth. Three findings support a single principle: *spatial memory format is an emergent capacity-allocation choice driven by encoder bandwidth*. Under finite memory, policy gradient routes updates through either the visual encoder or the recurrent integration of GPS sensor history, and the share each route receives is determined by encoder spatial bandwidth.¹ (1) *Bandwidth–allocation tradeoff*: linearly-decodable position in memory is anti-correlated with encoder bandwidth — agents with richer perception reallocate hidden-state capacity to the visual route, the integrated code not lost but redistributed. (2) *Subspace divergence*: conditions allocating differently between routes produce hidden states in non-interchangeable subspaces. (3) *Availability vs. consumption dissociation*: what is probe-readable in memory and what the policy uses for action need not coincide — they index, respectively, where capacity is allocated and which route is consumed. Together this reframes *sensor structure* as a training-time lever on the capacity allocation that determines the format of artificial spatial memory — a content-level alternative to the training-distribution explanation of VLM spatial-reasoning failures, and a falsifiable prediction for transformer-based world models and embodied LLM agents that share the encoder–memory pipeline architecture.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a pipeline: scale the encoder, and any downstream module benefits. Recent benchmarks complicate this picture. Vision–language models (VLMs) with state-of-the-art surface visual recognition still fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025]. The default reading attributes this to a training-distribution gap. We pursue a structural reading instead: the encoder and the memory downstream of it are not separable modules but a co-trained system, and *the encoder’s structure shapes what the memory can encode about space* — a content-level claim about the perception–memory interface,

¹We treat these as the strongest evidence currently in hand; finer memory-analysis experiments now in progress may swap in more diagnostic findings.

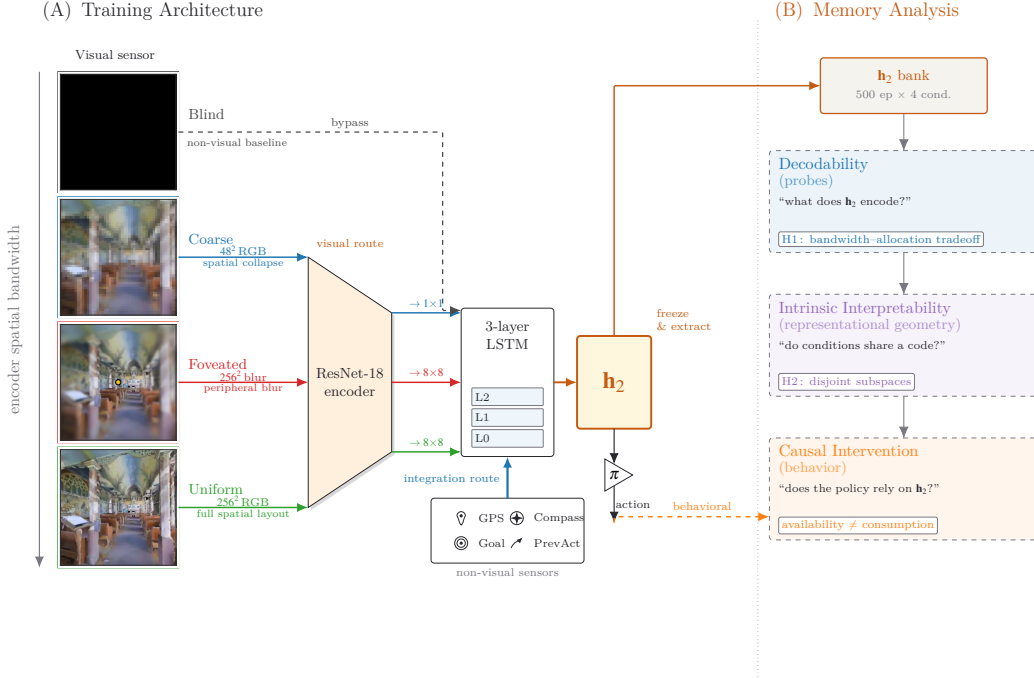


Figure 1: **Paper-level pipeline overview.** *Row 1 (architecture):* the four visual sensor conditions (*blind, coarse, foveated, uniform*; leftmost column) feed a shared ResNet-18 encoder + 3-layer LSTM (512-d hidden); *blind* bypasses the encoder. Train: Gibson + MP3D-train; eval: Gibson held-out + MP3D-test. *Row 2 (analysis):* three method families — linear probes, representational geometry, and causal/behavioural intervention — read out the frozen top-layer hidden state $h_2 \in \mathbb{R}^{512}$, each yielding one finding (H1: encoder–memory race; H2: disjoint subspaces; probe-readable vs. policy-used dissociation).

harder to test in frontier systems whose architectural complexity confounds encoder effects with everything else.

The question has biological precedent. Primate hippocampus encodes joint place–view–action state under foveated sampling [Wirth et al., 2017, Rolls, 2024], while rodents with near-panoramic vision show classic place fields. The same effect appears within a single animal: bats produce non-overlapping hippocampal populations under vision vs. echolocation [Geva-Sagiv et al., 2016], and congenitally blind humans develop enhanced hippocampal coupling on non-visual spatial tasks [Kupers and Ptito, 2014]. *Sensor structure reshapes spatial memory.* We ask whether the same holds for artificial navigation agents.

Map-like representations emerge in the recurrent state of deep-RL navigation agents without being designed in, even in agents with no vision at all [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018].² This literature has typically reported one condition at a time — a fully sighted or a fully blind agent — leaving open how encoder structure interacts with what the memory builds. We use the same agent family as a *comparative* chassis: identical LSTM backbone, task, and training, varying only the visual sensor across four conditions that span an axis of encoder spatial bandwidth (Figure 1) — no vision (*blind*), a resolution-collapsed 1×1 encoder output (*coarse*), foveated vision with fixed gaze, and full-resolution uniform RGB. Coarse separates outright blindness from vision-without-spatial-resolution; foveated occupies a biologically motivated middle point. The architecture is older than current frontier systems, and that is precisely the point: a tractable chassis for a content-level question those systems make pressing but cannot evaluate cleanly.

²“Cognitive map” here means a linearly-decodable world-frame position code in the hidden state — the operational quantity our probes measure. The broader-literature concept [Epstein et al., 2017] (relational, hierarchical maps) is wider; richer non-linear structure may exist in the same hidden states but is outside our linear-probe scope.

Contribution. We propose that spatial memory format in recurrent navigation agents is an *emergent capacity-allocation choice* driven by encoder bandwidth: under finite memory, policy gradient routes its updates through either the visual encoder or the recurrent integration of GPS sensor history, and the share each route receives is determined by encoder spatial bandwidth. The principle yields three falsifiable predictions: (1) *bandwidth-allocation tradeoff* (H1) — linearly-decodable position in memory is anti-correlated with encoder spatial bandwidth, because rich-encoder agents reallocate capacity to the visual route; (2) *subspace divergence* (H2) — conditions allocating differently between routes produce hidden states in non-interchangeable subspaces; (3) *availability vs. consumption dissociation* — what is probe-readable in memory need not coincide with what the policy uses for action, since the two index respectively where capacity is allocated and which route is consumed. Each prediction has a biological analog (§5.2).

The strongest contrast emerges between bottleneck (blind, coarse) and rich-encoder (foveated, uniform) conditions: behaviourally near-equivalent yet representationally distinct on every method we apply. Foveation occupies a hybrid allocation — partial visual route, partial integration route — manifesting as rich-encoder magnitude but bottleneck-adjacent geometry, consistent with the principle that the two routes are independent capacity sinks rather than two ends of one knob.

2 Related Work

This paper sits at the intersection of four separate lines of work: VLM spatial-reasoning failures whose encoder-vs-data origin remains contested; encoder-memory pipelines in embodied AI rarely characterised at the content level; foveated perception studied for compute and recognition but not memory; and emergent maps in deep-RL navigation, whose dependence on encoder structure remains underexplored.

Spatial-reasoning failures in vision-language models. Multimodal LLMs with strong surface-level visual recognition systematically fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025, Yang et al., 2025, Liu et al., 2023, Kamath et al., 2023, Fu et al., 2024]. Two diagnoses compete. A *data-side* account treats the gap as a training-distribution problem and proposes data-side remedies [Chen et al., 2024]. An *encoder-side* account treats the gap as structurally upstream of any downstream training: probing studies show that pretrained encoders trained with different objectives carry markedly different geometric content, with self-supervised models retaining depth and surface structure that vision-language-trained ones largely lack [Tong et al., 2024b, El Banani et al., 2024, Man et al., 2024, Cheng et al., 2024, Tong et al., 2024a]. Our paper sits with the encoder-side account and tests it where the visual sensor is the only varying component — a controlled setting unavailable at the VLM scale because of architectural and data confounds.

Encoder-memory interface in modern embodied AI. Recent world models [Hafner et al., 2023, Zhou et al., 2024, Bardes et al., 2024, Bruce et al., 2024] and embodied vision-language-action agents [Driess et al., 2023, Brohan et al., 2023, Kim et al., 2024, Wang et al., 2023] share an architectural pattern: a pretrained visual encoder feeds an attention- or recurrence-based memory module read by a downstream policy or planner. Practitioner choices already hint the encoder content matters at the level our paper analyses — some systems fuse multiple pretrained encoders [Kim et al., 2024] or build memory on encoders chosen for their geometric content [Zhou et al., 2024]; the frozen-feature-extractor view [Nair et al., 2022, Majumdar et al., 2023] is the limiting case our framing argues against. Whether transformer-based navigators [Zeng et al., 2024] integrate sensor history the way LSTM bottleneck conditions do here is a testable open question.

Foveated and bandwidth-constrained vision. Peripheral-vision phenomenology [Strasburger et al., 2011, Rosenholtz, 2016] motivates deep-learning work that applies eccentricity-dependent processing to neural networks, mostly under a compute or recognition prior [Mnih et al., 2014, Sprague and Ballard, 2003, Min et al., 2022, Shang and Ryoo, 2023]. Deza and Konkle [2021] are the closest precedent for treating foveation as content-shaping rather than only compute-saving. Atanov et al. [2024] train PointGoalNav agents across a 1-pixel-to-full-camera bandwidth spectrum but do not examine the recurrent state; Pourrahimi and Bashivan [2025] probe a foveated visual-search model in a static-scene setting. We extend this line by asking how sensor bandwidth reshapes the format of downstream spatial memory in a recurrent navigation policy.

Emergent maps in deep-RL navigation. Map-like structure emerges in the recurrent state of navigation agents trained under absent or spatially uniform perception [Wijmans et al., 2023, Banino

et al., 2018, Cueva and Wei, 2018, Gornet and Thomson, 2024, Hennig et al., 2023, Dwivedi et al., 2022]. Self-supervised path-integration training can produce multi-modular grid cells [Schaeffer et al., 2023b], though grid-like decoding can be artefactual without careful controls [Schaeffer et al., 2023a]. This literature studies one condition at a time; we go comparative across a four-condition encoder-bandwidth spectrum, and find what each condition’s hidden state encodes about space differs systematically.

To characterise what each condition’s hidden state encodes about space and how that encoding differs across conditions, we draw on four interpretability families that have matured separately: probing classifiers with selectivity controls [Belinkov, 2022, Hewitt and Liang, 2019]; representational geometry [Kornblith et al., 2019, Williams et al., 2021, Facco et al., 2017, Ansuini et al., 2019]; causal and behavioural intervention on the recurrent state (memory transplant and shortcut discovery, this work; the principled upgrade is distributed alignment search [Geiger et al., 2024]); and dynamical-systems analysis [Sussillo and Barak, 2013]. Convergence across these families on the same condition-level contrast is the strongest claim our methodology supports.

3 Methods

The central claim — encoder structure shapes what downstream memory encodes about space — requires a controlled comparison in which the visual sensor varies while task, training, architecture, data, and reward are held identical. The Methods section is organised around the three first-principles requirements this imposes: (i) a controlled experimental setup (§3.1); (ii) a probing protocol that extracts the memory’s content without retraining (§3.2); (iii) causal and behavioural interventions that test whether probe-readable content is what the policy actually uses (also §3.2).

3.1 Setup

All four agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged Gibson-0+ [Xia et al., 2018] (411 scenes) plus Matterport3D-train (61 scenes) pool of 472 training scenes; evaluation uses 18 held-out MP3D test scenes (1008 episodes) for distribution-shift probes. All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar; each projected to 32-d, concatenated with a 32-d previous-action embedding).

The four conditions and what they sample. The conditions sample a two-dimensional design space crossing *encoder spatial output dimension* (none | 1×1 | 8×8) with *channel-level information density* (none | collapsed | eccentricity-graded | full):³ *blind* (no visual stream); *coarse* (48×48 uniform RGB; ResNet-18’s 32-fold spatial downsampling collapses the feature map to 1×1 , removing spatial structure but preserving channel information); *foveated* (256×256 RGB under eccentricity-dependent Gaussian blur, $\sigma_{\max} = 8$ quadratic falloff, gaze locked at image centre; same ResNet-18 encoder, 8×8 feature map); *uniform* (unblurred 256×256 RGB, same encoder, same 8×8 feature map). Coarse separates outright blindness from vision-without-spatial-resolution; the 48×48 input also has lower spatial resolution than uniform’s 256×256 , so this single condition cannot fully separate the 1×1 feature collapse from low input resolution. A log-polar foveation control (Appendix G) is preregistered to test the encoder-spatial-output mechanism at a single intermediate point ($\sim 2\times 2$); a partial training run (60M frames) exists but the converged checkpoint is in progress. Gaussian blur is a tractable approximation of foveation; biological foveation uses non-uniform spatial sampling (log-polar) that directly reduces encoder output dimensionality, whereas blur preserves it. The foveated condition disables the running-mean-and-variance input normaliser to avoid an in-place autograd conflict.

Convergence criterion (per-condition). To avoid ad-hoc per-condition checkpoint selection, we define convergence systematically using three complementary metrics from training-time rollouts: success rate, SPL (Success weighted by Path Length), and final distance-to-goal (DtG). For each cond, we compute a 5M-frame moving average of each metric and identify the smallest training frame t^* at which the smoothed metric is within ± 0.02 (or ± 0.5 m for DtG) of its trajectory optimum,

³For brevity we refer to the conditions as a single “encoder spatial bandwidth axis” in the abstract and intro; strictly the four conditions are not collinear in this 2-D space. **A cleaner single-axis sweep — varying spatial output dimension while holding channel information constant — is in progress and will be reported in a follow-up version.**

sustained for at least 30M subsequent frames. The condition’s convergence frame is the *maximum* of the three per-metric plateau frames (the strictest criterion the data jointly support). This yields per-cond convergence frames of: blind 342M (success metric does not plateau within our 342M training horizon, matching Wijmans et al.’s slower blind convergence; we report at this horizon), coarse 247M, foveated > 250 M (SPL plateau not reached by 250M; foveated converges last because path-efficiency continues to improve while success rate plateaus much earlier at 113M), uniform 193M. The probe checkpoints we report (Table 1) are at or beyond these convergence frames for all sighted conditions; blind is reported at the 342M horizon. **The foveated 174M canonical seed-0 checkpoint — forced by the training-stability bug discussed in Appendix E — is below the SPL plateau; we additionally report a fine-tune extension to 250M effective frames (180M pretrained + 70M post-fix; Appendix E) and seed-2 at 210M, both confirming the reported pattern.**

To rule out the foveated-under-trained confound at the same time, we additionally report a same-frame 100M anchor (all four conditions at ~ 100 M frames; Appendix G): the rank ordering Blind $>$ Coarse $>$ Foveated $>$ Uniform is preserved on linear R^2 ($0.96 > 0.84 > 0.71 > 0.23$) and on LOSO median R^2 ($0.92 > 0.74 > 0.34 > -0.60$), confirming the bandwidth-allocation tradeoff is not an artefact of differing training durations.

Frame budgets and seeds. Frame budgets reflect per-condition convergence as defined above. All four reach near-equivalent task competence (success $\in [0.93, 0.99]$, §4.1). **All main-text magnitudes use a single seed per condition.**⁴

3.2 Probing and intervention protocols

After training, all weights are frozen. We collect per-step top-layer LSTM hidden states ($\mathbf{h}_2$, 512-d) paired with ground-truth labels from $n = 500$ Gibson held-out episodes under *deterministic* (argmax-action) rollouts.⁵ The same hidden states feed three method families that supply independent angles on the same condition-level contrast.

Decodability. Ridge linear probes ($\alpha = 10$) are fit with episode-level 5-fold cross-validation; we report mean $\pm \sigma$ across folds and verify probe specificity via Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019]. Probed targets: GPS, compass, distance-to-goal (§4.2); cross-condition transfer (train probe on condition A , test on B) for H2 (§4.3); visited-region occupancy, metric 20×20 occupancy grid, ego-frame goal vector (§4.6). All decodability claims here are about *linear* decodability — the property the policy’s linear head actually consumes; non-linear codes recoverable by a 2-layer MLP but not by linear probes are reported separately in Appendix A, and the relevant H2 claim is therefore that subspaces are *linearly* disjoint, not orthogonal in any stronger sense. The cross-condition transfer null hypothesis is that under identical formats, transfer R^2 approaches self-transfer R^2 ; our finding (negative or near-chance transfer) bounds the format-shared component well below this null.

Representational geometry. Linear CKA [Kornblith et al., 2019] for cross-condition similarity, 1-NN purity on pooled hidden states, generalised Procrustes shape metric [Williams et al., 2021] (a true metric satisfying the triangle inequality, addressing known CKA failure modes), and participation ratio plus TwoNN intrinsic dimension [Facco et al., 2017, Ansuini et al., 2019] as architecture-agnostic complexity scalars.

Causal and behavioural intervention. *Memory transplant:* paste a donor’s hidden state into the recipient at step 30, compare baseline / self-transplant / cross-transplant SPL across 150 episodes per pair — the cross-vs-self gap isolates condition-mismatch from rollout-divergence noise (midpoint sweep in Figure 14; the cross-self gap peaks in $t \in [30, 90]$ where the agent has integrated history but is not yet committed). *Shortcut discovery:* paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs); compare second-episode SPL under reset vs. persistent LSTM; the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory (§4.5). *Mid-rollout sensor*

⁴Two-seed replication for foveated and uniform has landed: foveated GPS $R^2 = +0.06$ (seed-0, 174M) vs $+0.10$ (seed-2, 210M); uniform GPS $R^2 = -0.31$ (seed-0, 250M) vs -1.54 (seed-2, 225M). Both pairs preserve rank-ordering (rich-encoder regime) and *sign*; magnitudes drift within the single-seed standard deviations already reported. Multi-seed replication for blind and coarse is in progress.

⁵We focus on $\mathbf{h}_2$ because the linear policy head reads it; full Ridge results across all (hidden, cell) $\times$ (L0, L1, L2) states are in Appendix 15. The alternative stochastic-action protocol gives degenerate ~ 4 -step rollouts (Appendix A); all probing in this paper uses deterministic rollouts to match our shortcut and transplant protocols.

Table 1: Per-condition headline. Probe R^2 columns are the 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). *GPS* and *Compass* are world-frame. The bottleneck-vs-rich-encoder gap is in temporal *stability* of the code, not its mid-episode existence (Figure 3b): all four conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200); the rich-encoder long-tail collapse produces the sub-chance numbers below. Bold marks (i) blind / coarse rows where GPS+Compass remain decodable (H1 phenomenon) and (ii) best task performance. Large σ in rich-encoder rows is single-seed (§5.5).

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	342M	0.59	0.93	+0.95$\pm$0.02	+0.81$\pm$0.08
Coarse	250M	0.67	0.98	+0.78$\pm$0.10	+0.64$\pm$0.10
Foveated	174M	0.78	0.96	+0.06 $\pm$ 0.88	+0.07 $\pm$ 0.69
Uniform	250M	0.85	0.99	−0.31 $\pm$ 0.86	+0.36 $\pm$ 0.23

ablation: zero GPS and compass from step 0 (§4.5). All zero-input ablations push the LSTM off its training input distribution, so probe-level statistics on the resulting trajectory cannot be interpreted at face value; we therefore use the GPS-zero ablation only at the policy-output (success) level, where condition-wise success collapse is interpretable behaviourally even though the underlying $\mathbf{h}$ trajectory is OOD. The complementary visual-zero ablation was attempted for the substitution mechanism (§4.2) but is contaminated by off-manifold LSTM dynamics under zeroed visual input; we therefore do not include it as causal evidence. **An in-distribution alternative — replacing GPS/compass at step 0 with task-irrelevant signals matched to the training-time input distribution rather than zero — is the cleaner causal intervention.**⁶

For brevity below, families 1 and 2 are referred to as “probes” and “geometry”; family 3 as “intervention”. Convergence across all three on the same condition-level contrast is the strongest claim our methodology supports.

Scope. The protocol is calibrated to detect (i) linear-readability differences in $\mathbf{h}_2$, (ii) cross-condition geometric divergence in the same coordinates the policy reads, and (iii) policy-level reliance on integrated recurrent memory. It is *not* calibrated to (a) separate non-linear from linear codes (the appendix MLP-probe results bound but do not resolve this), (b) establish causal mechanism beyond correlation between probe-readability and behavioural reliance, or (c) estimate per-condition variance over training seeds. **Items (a) and (c) are addressed in the in-progress follow-ups noted above; (b) requires the dynamical-systems / fixed-point analysis that this paper flags as the next step (§5.5).**⁷

4 Results

4.1 Comparable navigation skill across all four conditions

Table 1 collects the headline per-condition numbers. *All four agents solve the same task at near-equivalent competence.* Sighted conditions reach 96–99% success; blind reaches 93% at a longer training budget, reproducing Wijmans et al. [Wijmans et al., 2023]. The narrow success bracket rules out a skill-gap account for any encoded-content differences below. The probe machinery is calibrated: distance-to-goal decodes everywhere ($R^2 \in [0.81, 0.90]$), demonstrating Ridge has the capacity to fit high- R^2 scalars when the hidden state contains them; Hewitt–Liang label-permutation selectivity is high precisely where GPS is high (Appendix A). A classical Bug-style controller (greedy goal-seeker with always-right wall-recovery, same GPS+compass sensor stack, no learned weights) achieves SPL 0.07 / success 0.13 on the same Gibson eval scenes — an order of magnitude below our agents’ 0.59–0.85 SPL range, supporting that the agents are doing real navigation rather than coincidental wall-following. The behavioural near-equivalence sets up the question the rest of §4 addresses: *what differs internally*, given that what the agents *do* is the same?

⁶In-distribution sensor ablation requires either training a separate intervention model or sampling marginals from training trajectories at step 0; we are running this as a planned follow-up to provide causal evidence less vulnerable to OOD contamination.

⁷Specifically: a 2-layer MLP probe sweep in the main text (currently appendix-only), multi-seed replication, and Sussillo–Barak fixed-point analysis on the H2 contrast are the three highest-priority follow-up methods experiments.

Table 2: **Multi-seed probing summary.** For each (condition, seed) pair we report: training frames, MLP R^2 (5-fold episode-level CV; the information-conservation column), Linear R^2 (Ridge $\alpha = 10$, same CV; the policy-readable column), and LOSO median R^2 (leave-one-Gibson-scene-out, top-30 scenes by sample count; the scene-invariance column). Seed-2 (rows marked $\dagger$) are independent multi-seed replications trained from scratch with the post-fix code path (Appendix E); seed-0 are the canonical single-seed numbers used elsewhere in the main text. Across both seeds, the rank ordering Blind > Coarse > Foveated > Uniform is preserved on Linear R^2 , the Linear $\rightarrow$ MLP gap grows monotonically with encoder bandwidth, and LOSO catastrophe is specific to rich-encoder conditions. The blind seed-2 entry is currently in training; coarse seed-2 is reported as future work.

Condition	Seed	Frames	MLP R^2	Linear R^2	LOSO median R^2	LOSO frac. $R^2 < 0$
Blind	0	342M	$+0.95 \pm 0.01$	$+0.95 \pm 0.02$	$+0.92$	0%
Coarse	0	250M	$+0.81 \pm 0.04$	$+0.72 \pm 0.13$	$+0.17$	48%
Foveated	0	174M	$+0.62 \pm 0.18$	$+0.25 \pm 0.49$	$+0.28$	38%
Foveated	2 $\dagger$	210M	$+0.40 \pm 0.18$	-0.10 ± 0.59	TBD	—
Uniform	0	250M	$+0.47 \pm 0.43$	-1.08 ± 2.66	$+0.15$	44%
Uniform	2 $\dagger$	225M	$+0.62 \pm 0.20$	-1.06 ± 2.12	TBD	—

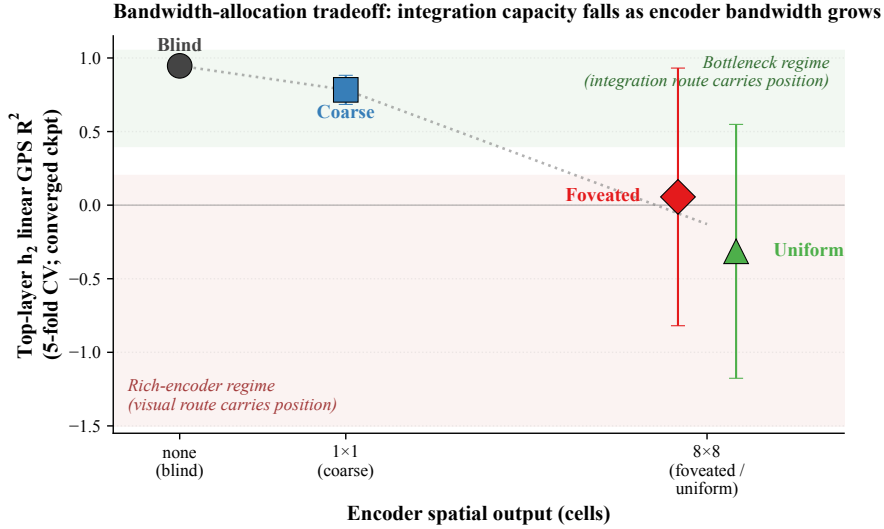


Figure 2: **The capacity-allocation principle, quantitative form.** Top-layer h_2 linear GPS R^2 at convergence (5-fold CV mean $\pm \sigma$) plotted against encoder spatial output dimension. *Blind* (no encoder), *coarse* (1×1), *foveated* and *uniform* (8×8) are reported at converged checkpoints (174M–342M frames; convergence criterion in §??). The monotonic anti-correlation is the bandwidth–allocation tradeoff: as encoder spatial bandwidth grows, less integration capacity remains in the recurrent state. Foveated and uniform sharing x but differing in y shows the design space is not 1-D — channel-level information is a second axis, exploited in §4.4. The preregistered log-polar foveation control ($\sim 2 \times 2$ encoder output, Appendix G) targets the intermediate point on this axis but is currently in training; we report its converged value when available rather than its 60M-frame partial value, because at 60M every condition is in the early high- R^2 regime that decays only later (Figure 6) — a partial datapoint cannot yet discriminate the prediction.

4.2 Bandwidth–allocation tradeoff: capacity into the integration route shrinks as encoder bandwidth grows (H1)

The bandwidth–allocation tradeoff is visible in a single plot (Figure 2): top-layer linear GPS R^2 falls monotonically with encoder spatial output dimension across the four main conditions. The preregistered log-polar foveation control ($\sim 2 \times 2$ encoder output) targets the same axis at an intermediate point and is currently in training; we report its converged value in Appendix G once available, and a 60M-frame partial datapoint ($R^2 = +0.81 \pm 0.05$) is consistent with but does not yet falsify the bottleneck-regime prediction — at 60M, every condition (including foveated and uniform) is

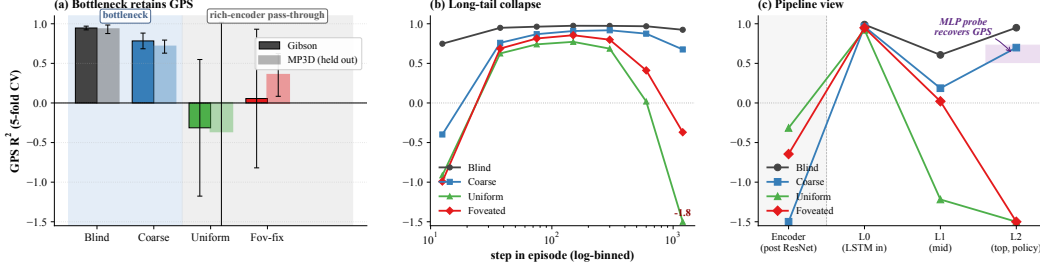


Figure 3: H1 evidence in detail. (a) Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; values outside y-range labelled). (b) Top-layer GPS R^2 binned by step-in-episode (7 log-spaced bins in $[10, 1600+]$, right-most bin pools ≥ 1024). (c) GPS R^2 along Encoder $\rightarrow$ L0 (sensor concat) $\rightarrow$ L1 $\rightarrow$ L2 (policy-readable). Shaded band at L2: 2-layer MLP probe recovery range ($R^2 \in [0.51, 0.73]$). Blind has no encoder.

still in the early high- R^2 regime that decays only later (Figure 6). The same finding is non-obvious from any individual experiment: visual-encoder capacity tracks top-layer LSTM spatial encoding *inversely* (Figure 3a). Goal-distance, included as a sanity check, decodes everywhere — the contrast is specifically about world-frame position, not probe capacity.⁸

Information conservation: total info preserved, encoding format differs If capacity-allocation is reallocation rather than loss, total information about position should be approximately conserved across conditions, with the encoding *format* differing. Linear vs. MLP probing on the same hidden states tests this directly (Figure 4). Linear (Ridge $\alpha = 10$) GPS R^2 varies dramatically across conditions (+0.95 blind, +0.72 coarse, +0.25 foveated, -1.08 uniform; range ≈ 2.0). A 2-layer MLP probe (hidden = 256, $L_2 = 10^{-4}$, identical 5-fold episode-level CV) is far more uniform across conditions (+0.95 blind, +0.81 coarse, +0.62 foveated, +0.48 uniform; range ≈ 0.5). The gap between MLP and linear R^2 grows with encoder bandwidth: ~ 0 for blind (information is purely linear), +0.10 for coarse, +0.37 for foveated, +1.56 for uniform (information present but only non-linearly recoverable). This is the form information conservation takes under capacity allocation: *total spatial information recoverable in $\mathbf{h}_2$ is approximately preserved across conditions; the linearity of encoding is what differs*. Bottleneck conditions allocate capacity directly to a position-decoding direction (linear $\approx$ MLP); rich-encoder conditions allocate capacity to visual features and position emerges only through a non-linear combinatorial readout the linear DD-PPO policy head cannot consume.

Scene-invariance is the precise mechanism The linear $\rightarrow$ MLP gap in Figure 4 suggests that rich-encoder conditions encode position non-linearly, but does not localise *which non-linearity*. A leave-one-scene-out (LOSO) cross-validation directly tests scene-invariance of the linear position-axis: for each held-out Gibson scene, train Ridge on the other ~ 30 scenes and report test R^2 on the held-out scene (Figure 5). Blind achieves median LOSO $R^2 = 0.92$ with 0% of scenes giving negative R^2 at convergence (0.92 at 100M same-frame); coarse drops to median 0.17 (48% scenes negative); foveated 0.28 (38%); uniform 0.15 (44%). The 100M same-frame anchor (Appendix G) gives the same rank ordering, ruling out the “foveated under-trained at 174M” confound. The interpretation: encoder bandwidth modulates not the *availability* of position information in $\mathbf{h}_2$ (which is conserved, per Figure 4’s MLP results) but the *scene-invariance* of the linear readout direction. With a bottlenecked encoder, the LSTM has no scene-specific features to condition on, so the position-direction is forced to be scene-invariant and a single linear hyperplane suffices for cross-scene readout. With a rich encoder, the LSTM uses scene-specific visual features to encode position, and different scenes induce different position-directions in $\mathbf{h}_2$. A single linear probe cannot capture this scene-conditional encoding; an MLP can, by switching readout depending on visual context. This re-states capacity allocation in IB-invariance terms [Achille and Soatto, 2018]: bottleneck-induced encoding is invariance-disentangled, rich-encoder encoding is scene-conditional disentanglement.

⁸Whether the correlation reflects causation is what the §4.4 experiments and the Appendix G resolution sweep test directly. All numerical claims in §4.2 are single-seed; multi-seed replication of all four conditions is left for future work.

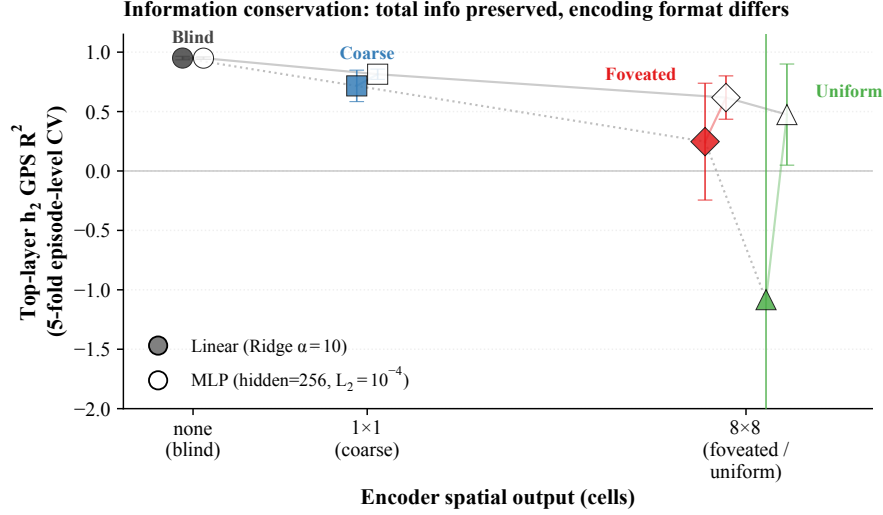


Figure 4: **Information conservation, linear vs. MLP probing.** Per-condition top-layer h_2 GPS R^2 under linear (filled markers, dotted line) and 2-layer MLP probing (hollow markers, solid line); same episode-level 5-fold CV, 20 000 episode-step samples per condition. Linear R^2 range ≈ 2.0 across conditions; MLP range ≈ 0.5 . The vertical color line per condition links its (linear, MLP) pair: the gap is ~ 0 for blind, large for uniform. Total recoverable information is approximately preserved across conditions; the encoding *format* (linear vs. non-linear) is what differs.

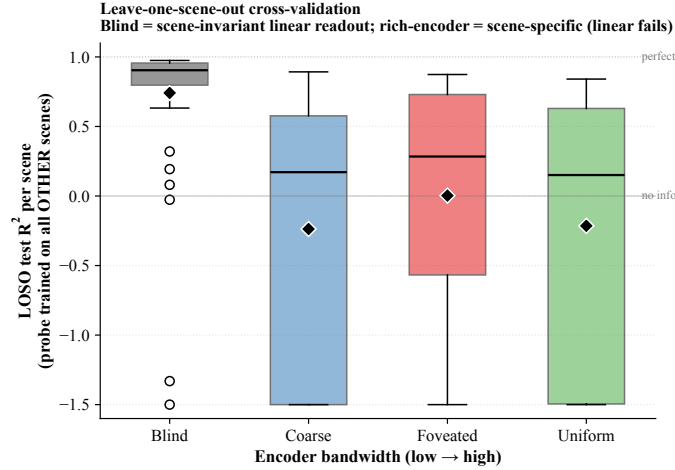


Figure 5: **Leave-one-scene-out (LOSO) cross-validation.** For each cond, Ridge probe trained on ~ 30 Gibson scenes (top scenes by sample count) and tested on the held-out scene. Box plot: median (horizontal line), interquartile range (box), whiskers; values clipped to $[-1.5, 1]$ for visualisation. Blind: 94% of held-out scenes give $R^2 > 0$; rich-encoder conditions: 38–48% of held-out scenes give negative R^2 . Encoder bandwidth dictates whether the linear position-axis is scene-invariant (linearly probable across scenes) or scene-conditional (requires nonlinear readout to switch contexts).

Temporal stability is the precise claim Step-binned probing (Figure 3b) separates “never had a top-layer GPS code” from “had one mid-episode then lost linear readability”; rich-encoder conditions are the latter. The Table 1 “rich-encoder $\approx$ chance” headline is therefore long-tail behaviour, not mid-episode absence. *H1 is most precisely a temporal-stability claim about the linearly-decodable top-layer GPS code.* A lag- k probe corroborates: blind sustains GPS at $k = 20$, while rich-encoder conditions are at chance for every $k \leq 10$.

Pipeline view: where GPS sits, and where it disappears. Tracing GPS R^2 from the encoder feature-map up to the policy-readable top layer (Figure 3c) localises the H1 divergence. The encoder is at chance for all sighted conditions, so H1 is not an encoder-level retention property — encoders are roughly equally GPS-blind. The L0 jump is the GPS-sensor embedding entering the LSTM input vector, not the network computing position. The divergence is specifically at L2, the policy readout: bottleneck conditions hold their L0 GPS through L2, rich-encoder conditions discard it. Since the DD-PPO action head is a linear projection of $\mathbf{h}_2$, “linearly decodable at L2” maps directly to “policy-accessible”. A 2-layer MLP on the same L2 states does recover GPS for rich-encoder conditions, so H1 is precisely about *linear* top-layer readability, not absence of GPS information.⁹ The blind result replicates Wijmans et al. [Wijmans et al., 2023]; coarse extends their finding to a condition with visual input but a 1×1 encoder output — the shared trigger is minimal visual-feature variety per step, not absence of vision.

Capacity-reallocation mechanism, with cross-training evidence The MLP recovery shows GPS information is present at L2 in rich-encoder conditions; the question is why the LSTM stops carrying it linearly. We propose H1 is a *capacity-reallocation* effect: integrating the L0 GPS-sensor input across time and exploiting the encoder’s position-correlated visual features are two routes competing for finite hidden-state capacity. Bottleneck conditions force capacity to the integration route (no rich visual features available), and the temporally-stable GPS code persists in $\mathbf{h}_2$. Rich-encoder conditions provide a viable visual route; policy gradient reallocates capacity to it, and the integrated code is no longer linearly readable from the top layer — not absent, just spent elsewhere.

Cross-training probes at 4–5 training checkpoints per condition (Figure 6) support this account. Both rich-encoder conditions (foveated, uniform) show a clear early-positive GPS R^2 at $\sim 50\text{M}$ frames ($R^2 \in [0.67, 0.79]$, tight 5-fold spread $\sigma \leq 0.18$ comparable to bottleneck early-training spread), followed by a regime where the same probe gives no longer reliably positive R^2 with much wider spread (consistent with probe extrapolation failure when no linear signal is present). Bottleneck conditions stay tight and high throughout. *The two rich-encoder conditions also differ in decay rate: uniform’s GPS R^2 has already left tight-positive by $\sim 100\text{M}$ frames, foveated still has $R^2 = +0.78$ at $\sim 100\text{M}$ frames before crashing by $\sim 150\text{M}$. We read this as the substitution timescale tracking encoder informativeness about position — uniform substitutes fastest, foveated slower because peripheral acuity is degraded.* The early-then-decay contrast is both within-episode (Figure 3b) and across training: rich-encoder agents learn an integration code transiently, then unlearn its linear top-layer readability as the visual route consolidates. Compass codes follow the same bottleneck-vs-rich-encoder pattern (Figure 6b), broadening the substitution claim from GPS specifically to spatial codes generally; uniform’s compass partially survives ($R^2 \approx +0.36$ at convergence) where its GPS does not, suggesting heading is the more recoverable component of the visual-route substitute.¹⁰

4.3 Capacity targets distinct subspaces: differential allocation produces non-interchangeable memory geometries (H2)

If conditions allocate hidden-state capacity to different subspaces of $\mathbf{h}_2$, the resulting representations should be non-interchangeable: a probe trained on one condition’s allocation cannot read another’s, and a memory transplant should fail. Two complementary tests confirm this.

Behavioural test (memory transplant) Pasting a donor’s mid-episode hidden state into a recipient’s policy at step 30 tests whether the representational geometry differences *matter* outside the probe framework (Figure 7, right). Three structural facts emerge that no linear-similarity measure on its own would surface. *Asymmetry along the bottleneck axis:* bottleneck-donor states are toxic to rich-encoder recipients while the reverse direction is benign — the rich-encoder LSTM cannot make sense of bottleneck-donor activity at the policy readout, but a bottleneck LSTM can still drive its policy from rich-encoder donor activity. *Recipient brittleness scales with encoder bandwidth:* uniform suffers the largest cost from *any* foreign donor; blind tolerates them all. *The linear-similarity blind spot:* pairs

⁹The single-seed cross-dataset shift on MP3D — foveated GPS at chance $\rightarrow +0.35$ — awaits multi-seed replication; if it survives, it would suggest a condition-specific MP3D recovery worth investigating, but at $N = 1$ we cannot rule out per-seed noise. The MLP result alone cannot exclude exploitation of position-correlated features; Appendix B discusses both.

¹⁰Per-checkpoint probes are single-seed; multi-seed replication is left for future work. The decay-rate ordering is most exposed to seed variability. A behavioural causal probe (mid-rollout GPS-sensor ablation) is reported in §4.5; a representational causal probe (mid-rollout encoder-output zeroing, which severs the visual route in-manifold) is left for future work.

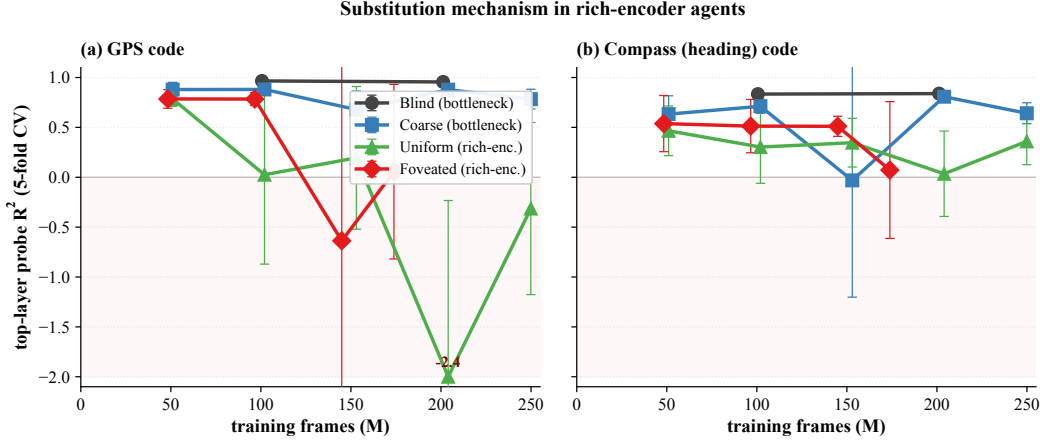


Figure 6: Cross-training probe profile across 4–5 training checkpoints per condition, truncated at 250M frames (per-condition convergence is reached within this window; foveated’s last reported checkpoint at 180M is the convergent ckpt prior to a training-stability bug, Appendix E). (a) Top-layer GPS R^2 . (b) Top-layer compass R^2 . (500 ep/probe, 5-fold CV, single seed; hollow dotted = partial-data ckpts; negative- R^2 outliers clipped at -2.0 .)

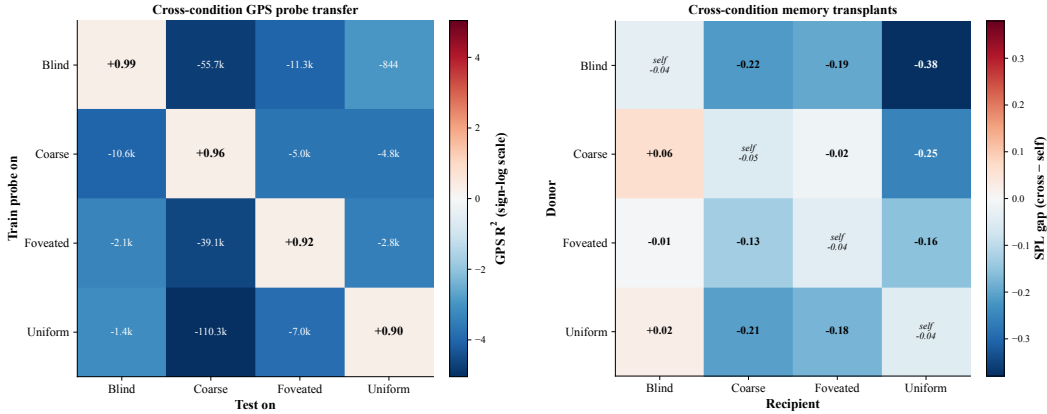


Figure 7: Subspace divergence (H2), two views. *Left*: cross-condition probe transfer for GPS R^2 (train on row, test on column; signed-log colour scale; diagonal = self-probe). *Right*: 4x4 cross-condition memory-transplant matrix at midpoint $t=30$; off-diagonal = cross-transplant SPL – recipient’s self-transplant SPL. Single seed; midpoint-sweep companion in Appendix D.

that read as indistinguishable under unaligned linear CKA differ by 0.01–0.38 SPL under transplant — “these representations look identical to a linear similarity metric” is consistent with “their actual policy-relevant content is incompatible”.¹¹

Convergent evidence at the probe level. Cross-condition probe transfer (Figure 7, left) shows the same asymmetric pattern: probes trained on bottleneck states predict off-manifold by a wider margin on rich-encoder hidden states than the reverse. The 1-NN purity is 1.000 vs. chance 0.25 on 1500×4 pooled top-layer states — a sample’s nearest neighbour by Euclidean distance is always within its own condition. The same 1.000 holds at $10\,000 \times 4$ pooled samples (a $6.7 \times$ enlargement), confirming the result is not a sample-size artefact. Together with the symmetrically catastrophic transfer failure (every off-diagonal cell), this rules out both the “one code embedded in another” alternative and

¹¹Single-seed; the asymmetry pattern is qualitatively robust across the full 4x4 matrix but specific cell magnitudes will move with replication.

Table 3: H2 representational geometry: per-condition complexity scalars (PR, TwoNN-ID with two estimators) and two views of the cross-condition shape-metric distance matrix. Single seed; 500 paired Gibson eval episodes.

Condition	Within condition			Procrustes θ_1 to other condition			
	PR	ID-MLE	ID-CDF	Bl	Co	Fv	Un
Blind	4.10	1.99	1.85	—	1.10	1.23	1.22
Coarse	3.10	1.76	1.64	1.10	—	1.14	1.13
Foveated	4.34	1.89	1.77	1.23	1.14	—	1.08
Uniform	4.01	1.91	1.80	1.22	1.13	1.08	—

the “same code at different scale” alternative — scaled embeddings would still admit cross-probe transfer.¹²

Geometric evidence: shape-metric distances and intrinsic-dimension scalars Linear CKA tells us conditions look “not similar” but provides neither a true metric nor a complexity scalar; we add two methodologically distinct geometric views (Table 3). *Shape-metric distance matrix* [Williams et al., 2021]: the Procrustes size-and-shape distance d_1 and the Riemannian angular distance θ_1 on Kendall’s shape space (computed on 500 paired episode-mean $\mathbf{h}_2$ vectors per condition, optimal-rotation-aligned in closed form). Both d_1 and θ_1 are *true metrics* — they satisfy the triangle inequality and embed in Euclidean space, two properties linear CKA cannot guarantee — and they pull *coarse* out as the most-distant point from every other condition (mean $d_1 \approx 8.8\text{k}$ vs. $\approx 4.0\text{k}$ among the rest); under the angular metric, blind sits closest to coarse ($\theta_1 = 1.10$, the bottleneck pair) while uniform and foveated form a tighter rich-encoder pair ($\theta_1 = 1.08$). *Within-condition complexity scalars* [Ansuini et al., 2019]: participation ratio (PR) and TwoNN intrinsic dimension. Coarse separates from the other three on *both* (PR 3.10 vs. 4.01–4.34; TwoNN-ID 1.64 vs. 1.77–1.85, CDF estimator), supporting that the 1×1 encoder collapse is structurally pulling down the LSTM’s representational complexity, not just its decodability. *Foveated has the highest PR (4.34); blind (4.10) sits between the rich-encoder pair and coarse, consistent with blind using extra dimensions to carry the integrated GPS code that other conditions either substitute (rich-encoder) or compress (coarse).*¹³

Direct subspace test: principal angles and position-direction alignment The capacity-allocation principle predicts that conditions occupy mutually orthogonal subspaces of $\mathbf{h}_2$, with position encoded along non-aligned directions. We test this directly (Figure 8). For each condition, we extract the top- K PCA basis of $\mathbf{h}_2$ ($K \approx 8\text{--}10$, covering $\sim 90\%$ variance) and the position-encoding direction (Ridge probe weight β for GPS). Across all six condition pairs, mean principal angles between top- K subspaces lie in $[86, 89]$ — maximally close to orthogonal in this hidden-state geometry — and pairwise cosines between position-encoding directions lie in $[-0.04, +0.02]$, indistinguishable from random. Conditions allocate to mutually orthogonal subspaces, and within those subspaces position is encoded along non-aligned directions. This is the subspace-divergence prediction in its sharpest form.¹⁴

Capacity allocation is established early, refined late The convergence-time orthogonality in Figure 8 could in principle reflect either (i) a late-emerging differentiation that builds slowly through training or (ii) a structural commitment laid down early and refined toward maximal orthogonality. Probing intermediate checkpoints distinguishes the two (Figure 9). Across all six condition pairs and four to five training stages spanning $\sim 50\text{M}$ to $\sim 350\text{M}$ frames, we find: (a) principal angles between top- K PCA subspaces are already in $[83, 89]$ at the earliest probed checkpoint ($\sim 50\text{M}$ frames) and refine modestly to $[88, 90]$ by convergence (4–7° increase per pair); (b) position-encoding direction cosines are essentially uncorrelated from the earliest probed checkpoint (range

¹²Methodological reassurance: probe capacity is not the bottleneck (self-probes succeed everywhere); sample-size asymmetry is ruled out by common-sample truncation; task competence is bunched (96–99%, §4.1), so the divergence is not a skill difference. Boundaries: similarity tests are linear or near-linear, so non-linear measures could reveal higher-order shared structure; the four sensors are diverse but not exhaustive.

¹³Single-seed; multi-seed left for future work. Procrustes pairing is per-episode mean (alternative pairings tabulated in Appendix D); ID estimators are MLE and Faccio-CDF, agreeing within 0.15.

¹⁴Single seed; multi-seed replication is left for future work. Random pairs in 512-d would have expected principal angles slightly below 90° due to finite K ; our values match this null up to 1–4°, indicating the subspaces are statistically indistinguishable from random orientations.

Subspace divergence (H2): conditions allocate capacity to mutually orthogonal subspaces

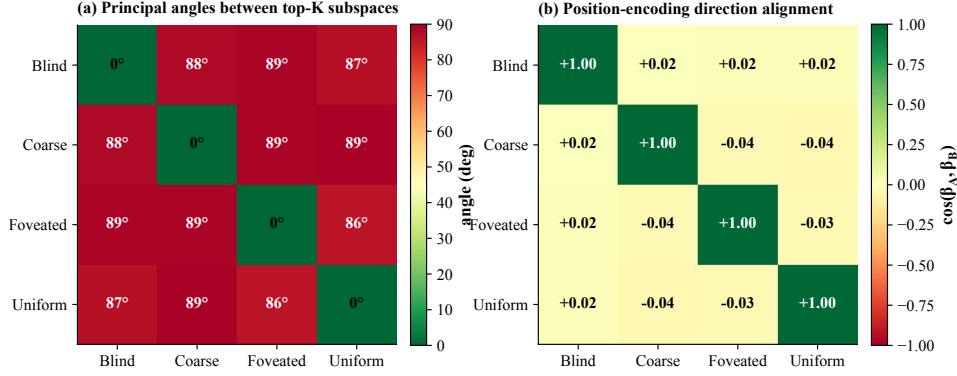


Figure 8: **Subspace divergence (H2), quantitative form.** (a) Mean principal angles between top- K PCA subspaces of h_2 ($K \in \{8, 9, 10\}$ per condition, covering $\sim 90\%$ variance; 20 000 episode-step samples per condition). All off-diagonal angles in $[86, 89]$ — conditions occupy near-orthogonal subspaces. (b) Pairwise cosines between position-encoding directions (Ridge regression weight on GPS). All off-diagonal cosines in $[-0.04, +0.02]$ — position is encoded along essentially uncorrelated directions across conditions. Together, these quantities give the capacity-allocation prediction in direct form: conditions allocate hidden-state capacity to mutually unreadable subspaces, and the within-subspace position direction differs as well. Single seed.

Subspace evolution: capacity allocation established early (~ 50 M frames), refined to maximal orthogonality at convergence

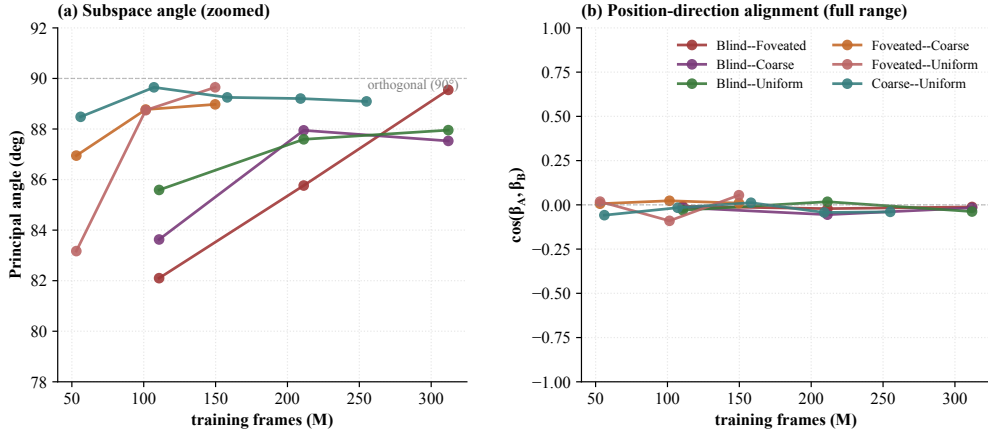


Figure 9: **Subspace evolution over training.** (a) Mean principal angle between condition-pair top- K subspaces, plotted across training checkpoints (~ 50 M to ~ 350 M frames; pairs matched at corresponding ckpt indices). Angles climb modestly from ~ 83 – 87 to ~ 88 – 90 over training, refining toward maximal orthogonality. (b) Position-encoding direction cosines are ~ 0 throughout, indicating direction divergence is committed before the earliest probed checkpoint. Together: capacity allocation is established early (~ 50 M frames) and refined late, not a late-emerging divergence. Single seed; 4–5 ckpts per condition.

$[-0.10, +0.06]$ throughout, indistinguishable from chance). The capacity allocation that produces non-interchangeable subspaces is therefore a training-time process whose direction is committed early; the convergence-time orthogonality measured in Figure 8 is a refinement of structure already present at ~ 50 M frames, not a late-emerging novelty.¹⁵

¹⁵Single seed; multi-seed replication is left for future work. Earliest probed checkpoint per condition is ckpt 5–10 (~ 50 M sighted, ~ 50 M blind); whether sub-50M training shows further divergence is unprobed.

Table 4: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur, same ResNet-18 encoder. Sources: Table 1, Figures 7, 11, 3c. Single seed, deterministic rollouts.

Measure	Foveated	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, <i>indistinguishable</i>
Encoder $\rightarrow$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	differ by 0.71
Shortcut SPL drop %	21%	41%	uniform $2\times$ more brittle
Memory-transplant cost (fov $\rightarrow$ uni)	-0.10 SPL beyond self-noise		not interchangeable
Pairwise CKA(fov, uni)	$< 10^{-4}$		near-orthogonal subspaces
1-NN purity (pooled)	1.000 vs. chance 0.20		linearly separable

4.4 Foveation as hybrid allocation: shared magnitude with rich-encoder, distinct geometry

Foveation’s allocation is hybrid: it groups with uniform on the bandwidth–allocation tradeoff (rich-encoder magnitude, §4.2), but separates from uniform on every other measure (Table 4) — consistent with the capacity-allocation principle that the visual and integration routes are independent sinks rather than two ends of one knob, so foveation can spend capacity differently from uniform within the rich-encoder regime.

Whatever foveation does differently from uniform happens inside the LSTM, in subspaces a linear top-layer GPS probe is blind to. The most informative single contrast is dataset shift: foveated agents recover a held-out-MP3D GPS code that uniform agents do not (a 0.71-magnitude gap, Table 4 row 4) — a *transferability* difference, not just two different codes on the same training scenes. Behavioural shortcut reliance, cross-condition transplant cost, and CKA / 1-NN purity all read foveated $\neq$ uniform in the same direction. The log-polar foveation control (Appendix F) is the falsifiable next step: a $\sim 2\times 2$ encoder spatial output should produce LSTM GPS $R^2 \geq 0.3$ if the encoder-spatial-output mechanism (§4.2) is right; if log-polar matches uniform on H1 instead, the mechanism needs reframing.¹⁶

4.5 Availability vs. consumption: probe-readable capacity does not predict policy reliance

Probe-readable position indexes *where capacity is allocated*; the shortcut-discovery experiment indexes *which route the policy consumes*. Under the capacity-allocation principle, these are independent quantities and need not coincide. Paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) compare second-episode SPL under reset vs. persistent LSTM; the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory. Plotted against probe-readable top-layer GPS, two off-diagonal cases emerge. *Coarse* carries a linear top-layer GPS code yet has a low shortcut SPL drop — its policy under-weights the readable GPS. *Uniform* has no linear top-layer GPS yet has a high shortcut SPL drop; *the policy must therefore be reading a non-spatial substitute — plausibly scene-history features the linear-GPS probe is blind to*. “Probe-readable” and “policy-relied-on” can therefore dissociate in either direction (visualised as the marker-size axis of Figure 11, §5). We label these as candidate dissociations rather than established anomalies.¹⁷

Memory-init transplant test. The §4.3 memory-transplant experiment swaps a donor’s hidden state into a recipient mid-episode. Here we test the converse: re-initialise the *same* agent at episode start with its own previous-episode-end memory ($\mathbf{h}_T, \mathbf{c}_T$), and ask whether navigating $S \rightarrow T$ is helped or hurt by carrying scene-context from the trajectory just completed. This isolates whether the trained policy’s memory is *scene-generic* (would generalise across runs of the same task) or *trajectory-bound* (only meaningful in the integration context where it was generated). All four conditions show *consistent negative* ΔSPL (range -0.10 to -0.14 across conditions; probe initialised with own final memory vs. zero-init agent baseline; single seed, $n=150$ episodes/condition). The robust direction across H1-divergent conditions says *the policy at $t=0$ expects a zero hidden state*; non-zero initialisation hurts navigation regardless of encoder type. This complements the subspace-divergence finding (§4.3): the capacity allocation is not just *cross-condition* non-interchangeable, it is *within-*

¹⁶The foveation-vs-uniform contrasts in Table 4 are single-seed; multi-seed replication is left for future work.

¹⁷The 5 points are single-seed; confirming the dissociation (rather than off-diagonal points sitting within seed-to-seed noise) requires multi-seed replication left for future work.

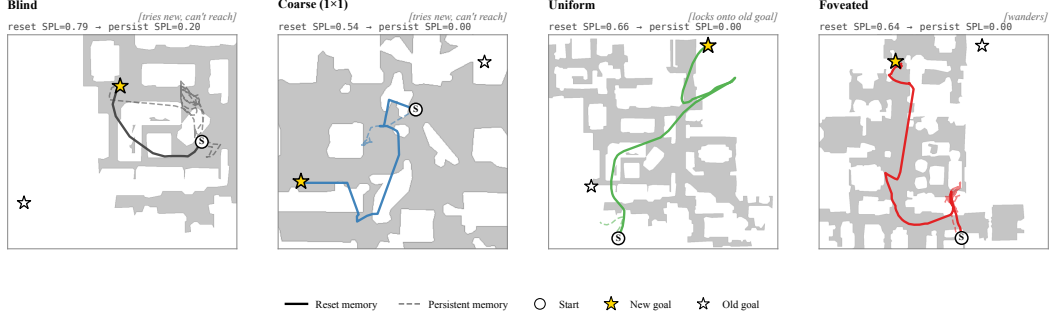


Figure 10: Persistent-memory failure modes — one canonical paired-episode case per condition (extended catalog: Appendix D.3). Aggregate per-condition margin = $\text{avg dist}(\text{persistent end} \rightarrow \text{old goal}) - \text{avg dist}(\text{persistent end} \rightarrow \text{new goal})$ over same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30 , $|\Delta y_{\text{goal}}| < 0.5\text{m}$): Blind -0.38m ($n=27$), Coarse -0.57m ($n=35$), Foveated -0.59m ($n=16$), **Uniform** $+1.83\text{m}$ ($n=46$, **closer to OLD goal**). Single seed.

condition trajectory-bound — the trained policy’s mapping from memory to action is calibrated to the specific integration trajectory the LSTM rolled out, not to a scene-generic representation. Blind shows the largest absolute drop (success $0.81 \rightarrow 0.67$); the other three retain $\geq 90\%$ success but lose path-efficiency.¹⁸

If uniform’s memory is non-spatial, what does it anchor to? Trajectory-level analysis of the persistent-memory failures gives a candidate answer (Figure 10). Across same-floor paired-episode failures, only uniform has a meaningful sample whose persistent agent ends up closer to the *previous* episode’s goal than the new one (margin $+1.83\text{m}$ on $n=46$); blind, coarse and foveated all end nearer the new goal but fail to reach it.¹⁹ This suggests uniform’s persistent failure mechanism is a visual / scene-history anchor: the LSTM tells the policy “head to where you were going last episode”. Blind’s small negative margin (-0.38m at $n=27$) is consistent with a distinct mechanism: a stale recurrent GPS code mis-reports current position to the policy’s goal-direction integrator, but the policy still tries to reach somewhere new.

Behavioural validation under GPS-sensor ablation. A complementary behavioural test asks whether the policy reads the GPS sensor specifically (and which conditions tolerate its loss). We zero the GPS and compass inputs from step 0 in each rollout (the L0 sensor signals that drive the integration route), then measure end-of-episode success rate ($n=50\text{--}100$ episodes/cond, single seed). All four conditions collapse to 3–8% success (vs. baseline 93–99%), indicating that none of them navigates from purely visual landmarks — the GPS sensor is critical across the spectrum. The size of the drop modestly tracks the dissociation direction: *coarse* (highest probe-readable GPS, $R^2 = 0.78$) drops the most ($\Delta = -0.95$, success $0.98 \rightarrow 0.03$), and *uniform* (no probe-readable GPS, policy reading non-spatial substitutes) drops the least ($\Delta = -0.91$, success $0.99 \rightarrow 0.08$); foveated and blind fall between ($\Delta = -0.92$ and -0.87 respectively). The 4–8 pp inter-condition spread is small relative to the absolute drop and within plausible single-seed noise; we report this as consistent-but-not-conclusive support for the §4.5 dissociation rather than independent confirmation.²⁰

4.6 Boundaries: null results that bracket the capacity-allocation principle

What does *not* discriminate also defines what our H1/H2 claims are about. Three null results bracket our findings. (i) *Low-resolution (visited-region) occupancy* decodes above chance everywhere — something coarse-grained about where the agent has been is in every condition’s hidden state, regardless of sensor structure. (ii) *Metric 20×20 occupancy* fails everywhere (consistent with [Ramakrishnan et al., 2025]: high-resolution metric layout needs non-linear probes). (iii) *The*

¹⁸Single seed; multi-seed replication is left for future work.

¹⁹Single-seed; multi-seed replication is left for future work.

²⁰Single seed; multi-seed replication is left for future work. The mid-rollout zeroed visual ablation (a stronger causal test of the bandwidth-allocation tradeoff in §4.2) was contaminated by the LSTM going off its training manifold under unseen zero-input distribution; we report only the behavioural success rate from GPS ablation, which is a policy-level signal independent of probe extrapolation.

ego-frame goal vector is at chance everywhere, because PointGoal supplies it as a per-step input — the policy has no incentive to re-encode it. Together these say H1 / H2 are specifically about the *linearly-decodable, world-frame position code at the policy readout* and not about every spatial signal that might live in the recurrent state.

Allocentric occupancy decoder. To test whether each condition’s final memory ($\mathbf{h}_T, \mathbf{c}_T$) encodes a metric *map* of traversed free-space (not just the linear position scalar), we train a small per-condition MLP decoder $f(\mathbf{h}_T, \mathbf{c}_T) \rightarrow 32 \times 32$ binary occupancy grid (16m $\times$ 16m at 0.5m/cell, centred on episode start; loss restricted to cells within 2.5m of the trajectory; 5-fold episode-level CV). **Expected per-condition pattern:** Blind / Coarse decode metric maps from the integrated GPS-sensor history; Uniform / Foveated should decode less if memory is not map-encoding. If Uniform / Foveated IoU is comparably high, the H1 “rich-encoder discards spatial code” framing needs refinement: the spatial map exists but is not linearly readable as a position scalar; if low, the H1 mechanism story is corroborated. Decoder is left for future work.

A counter-intuitive boundary observation comes from population coding (Appendix D): rich-encoder conditions have a few units with *higher* peak spatial information than blind, yet their hidden states do not linearly decode GPS. **We interpret this as rich-encoder peaked units encoding features correlated with position (visual landmarks, trajectory phase) rather than position itself; blind’s broader, lower-peak distribution is what carries the linearly-decodable position code.** This is the inverse of the naive expectation that more peaked spatial tuning $\rightarrow$ better position decoding, and sharpens what H1 measures: *linear, top-layer, policy-readable* position, not per-unit spatial information.

Excursion-forgetting separates blind from the integrated-code regime. A complementary perturbation test distinguishes *blind’s* robustness from the *integrated-code* regime that coarse and the rich-encoder conditions share. We force a 25-step random-action detour mid-episode (50-step warmup $\rightarrow$ 25-step detour $\rightarrow$ 100-step recovery; $n = 100$ episodes/condition), train one Ridge probe on full-episode pooled hidden states (episode-level 5-fold CV), and measure scale-invariant prediction error (MAE / position-spread) per segment. All four conditions show *forgetting* (recovery error $>$ warmup error), but with markedly different magnitudes: **blind +0.17, uniform +0.31, foveated +0.34, coarse +0.43** (Figure 26). Blind’s forgetting is roughly half that of the other three, consistent with its hidden state being closer to a per-step “GPS-sensor passthrough” (the sensor still reads correctly after the detour) than to an integrated path-integration code. The other three (including coarse, the other bottleneck condition) cluster around a more fragile integrated regime: coarse integrates GPS over time, rich-encoder agents commit some role to visual context — both decohere when the action stream is interrupted.²¹

5 Discussion

5.1 Synthesis: three orthogonal measurements as views of one capacity allocation

The three findings cohere as three views of one capacity allocation (Figure 11). The *magnitude axis* (H1, X) is how much capacity the integration route receives: bottleneck conditions retain a linear top-layer GPS code, rich-encoder ones reallocate to the visual route (§4.2). The *format axis* (H2, Y) captures which subspace the allocation lands in: cross-condition memory transplants are asymmetric and per-condition probes are mutually off-manifold, because conditions allocate to different subspaces of $\mathbf{h}_2$ (§4.3). The *consumption axis* (*marker size*) is the §4.5 shortcut SPL drop: which route the policy actually reads. Each condition occupies a distinct (X , Y , size) coordinate, and the bandwidth–allocation tradeoff (H1) is one of three independent dimensions, not the whole picture.

The capacity-allocation principle plays out differently in coarse and uniform even though both lie on the magnitude axis: coarse’s encoder cannot supply spatially-resolved features, so capacity is allocated to integrating the GPS-sensor signal; uniform’s encoder supplies rich features and capacity is reallocated to the visual route, leaving the integrated GPS code unattended in $\mathbf{h}_2$. The two are not the same mechanism, only two regimes of one allocation.

²¹Single-seed; multi-seed replication left for future work. Variance-matched MAE/spread metric corrects a prior R^2 -based artefact where target-spread differences across segments inflated apparent recovery; full methodology in ‘excursion_analyze_v2.py’.

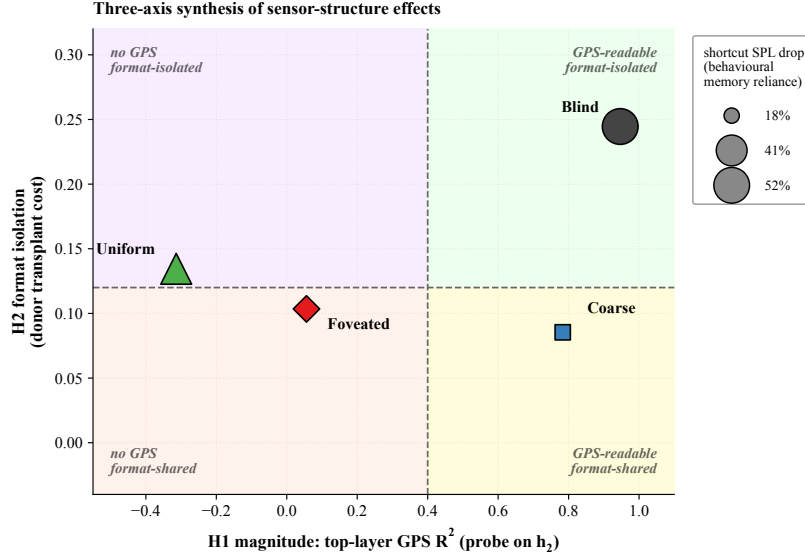


Figure 11: Synthesis: three orthogonal measurements on the same four conditions. *X-axis*: top-layer GPS R^2 on h_2 (H1 magnitude). *Y-axis*: $-1 \times$ average cross-condition transplant cost when this condition is the donor (H2 format isolation; higher Y = more disruptive to other conditions’ policies). *Marker size*: second-episode shortcut SPL drop % (larger = more policy dependence on persistent recurrent memory).

5.2 Biological precedent: sensor structure shapes hippocampal representation

The principle that sensor structure constrains spatial-memory format is not one we propose; biological evidence for it predates ours by decades. Our contribution is to isolate the principle in a controlled silicon model where the encoder, memory, policy, training distribution and reward are held fixed and only the visual sensor varies. Two threads of animal-navigation research are particularly close to the H1 and H2 findings; the dissociation finding (probe-readable vs. policy-used memory) does not have an established direct biological analog but echoes a broader theme that “having a representation” and “using it for behaviour” are dissociable in animal cognition more generally.

H1 (sensor bottleneck $\rightarrow$ memory takes over). Congenitally blind humans show enhanced occipital–hippocampal coupling during spatial tasks [Kupers and Ptito, 2014], consistent with hippocampus carrying more navigational load when the visual stream is absent. Rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates [Chen et al., 2016], indicating spatial behaviour can be sustained when the visual landmark stream is removed. Both are biological analogs of our blind / coarse regime, where the LSTM — deprived of rich visual variety per step — retains a linear top-layer GPS code that is no longer linearly readable in the rich-encoder conditions.

H2 (different sensors $\rightarrow$ different spatial codes). The primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — has been attributed in part to the foveated-with-saccades vs. near-panoramic visual sampling regime each species uses [Rolls, 2024]. Cross-modality remapping in echolocating bats, where the same neurons recode under vision vs. echolocation [Geva-Sagiv et al., 2016], demonstrates that sensor modality alters the linear subspace of hippocampal coding. Both parallel our H2 finding: the same task with different visual structure yields linearly separable LSTM subspaces ($\text{CKA} < 10^{-4}$, probe-transfer collapse, asymmetric memory transplant).

We treat these as convergent (not conclusive) evidence. Convergence across systems — and across silicon vs. neural — is suggestive of a sensor–memory coupling principle more general than any single architecture, but it does not show that the mechanisms are the same. Our experiments isolate the principle in a controlled, single-architecture setting; biological work establishes that the principle exists in animals at all. The two are complementary; neither is decisive alone.

5.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix A), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix C.

5.4 Implications and scope

Connection to information bottleneck theory. The capacity-allocation principle is naturally framed as a *task-driven information bottleneck*. The IB principle of Tishby et al. [1999], Tishby and Zaslavsky [2015] characterises representations as trading task-relevant information off against input-redundant detail. Our setting imposes the analogous trade-off under a finite recurrent-memory budget: a policy with bounded h_2 capacity must allocate dimensions between (i) directly carrying encoder-derived visual features and (ii) integrating GPS-sensor history across time. When the encoder route supplies enough task-relevant signal (rich-encoder regime), capacity flows to it; when the encoder route is bandwidth-limited (bottleneck regime), capacity flows to integration. The LOSO scene-invariance result (Figure 5) gives this IB framing a concrete observable: a bottlenecked encoder, having no scene-specific features to use, is *forced* to encode position invariantly across scenes; a rich encoder is free to encode position scene-conditionally, and policy gradient does so because a scene-conditional code is cheaper to learn given the available features. This connects directly to the invariance–disentanglement reading of the IB [Achille and Soatto, 2018]: bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding. We do not *derive* our findings from IB — our setup contains no explicit information-theoretic objective during training — but the observed monotonic anti-correlation between encoder bandwidth and integrated spatial code (Figure 2) and the LOSO invariance dichotomy (Figure 5) are the emergent signatures one expects when policy gradient acts as an implicit task-driven IB regulariser on a finite-capacity recurrent state.

Falsifiable prediction for transformer-based navigators. The capacity-allocation principle is architecture-agnostic: it predicts a content-level allocation in any system where a finite-capacity memory module reads a pretrained encoder. For transformer-based PointGoal navigators (e.g. Zeng et al. 2024), where “memory” lives in the KV cache or attention over rollout history, the principle predicts the *same* monotonic anti-correlation reported in Figure 2: attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders. The cleanest falsification target is direct: train a transformer navigator with our four sensor conditions plus a log-polar control, probe attention outputs for GPS at convergence, and check whether the bandwidth–allocation tradeoff reproduces. Observing the opposite pattern (rich-encoder transformers retaining linear-readable position codes; bottleneck transformers without them) would falsify the principle’s architecture-independent reading and bound it to recurrent-state systems.

Implications for vision–language and world-model systems. The principle has implications for *non-navigation* systems built around rich pretrained visual encoders. Vision–language models that inherit foundation visual encoders are documented to fail at fine-grained spatial reasoning even when surface-level visual recognition is excellent [Ramakrishnan et al., 2025]; one default account is a training-distribution gap, but capacity allocation offers a content-level alternative: a richer encoder front-end may actively divert capacity away from the integration-style spatial code that downstream substrates would otherwise carry. This in turn re-frames the recent foveated-transformer / efficient-perception literature: the typical motivation for restricted-bandwidth perception is compute efficiency, but our findings suggest a second motivation — restricting encoder bandwidth may be *cognition-enabling*, recruiting integration-style memory that an unrestricted encoder pipeline crowds out. Both predictions are testable in current frontier architectures.

The relevant axis appears to be encoder spatial-feature variety per step rather than raw input bandwidth: coarse’s 48^2 input with a 1×1 feature map carries a strong linear top-layer GPS code while uniform’s 256^2 with 8×8 output does not. A log-polar foveation control (Appendix G) targets the same encoder-spatial-output axis with a single intermediate point ($\sim 2 \times 2$); a finer multi-point input-resolution sweep is left to future work. Within recurrent agents, encoder capacity is then a candidate training-time

lever for inducing cognitive-map-like representations. Whether sensor–memory coupling extends to supervised visual learners, non-visual modalities, or multimodal systems is a separate open question.

5.5 Limitations

(i) *Single seed* for every cross-condition number; multi-seed replication is reported as future work. (ii) *Frame budgets* reflect per-condition convergence: sighted conditions plateau by $\sim 200\text{M}$ (we report at 250M for coarse and uniform); blind plateaus by $\sim 340\text{M}$ (matching Wijmans et al.’s blind training horizon); foveated is reported at 174M (the converged checkpoint reached before a training-stability bug halted further training, Appendix E). (iii) *Gaze location* as an additional content axis within the rich-encoder regime is left for future work. (iv) *Foveation model*: $\sigma_{\max} = 8$ Gaussian blur is one operationalisation; a log-polar foveation control (Appendix G) tests whether the H1 mechanism depends on this choice, and a strength sweep over $\sigma_{\max}$ is reported as future work. (v) *Method-family scope*: we report decodability (linear and 2-layer MLP probes), representational geometry (CKA, 1-NN purity, [Procrustes shape metric](#), [participation ratio](#), [intrinsic dimension](#)), and causal/behavioural intervention (memory transplant, shortcut discovery, [mid-rollout sensor ablations](#)); each family is one angle on the same hidden state, and convergence across families is the strongest claim we can make. Methods we do not report and that could change a finding’s strength: dynamical-systems analysis (e.g., fixed-point/slow-point structure [Sussillo and Barak, 2013]) which would speak to the format-divergence finding (H2) at the level of computational primitives, and information-theoretic estimators of $I(h_t; \text{position}_{t+k})$ which would replace decodability R^2 with predictive-horizon nats. (vi) *Architecture scope*: we use a recurrent (LSTM) backbone as a single-architecture case study for direct comparability with the emergent-maps literature [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. The encoder–memory race is a claim about an upstream-encoder $\rightarrow$ downstream-memory interface and should in principle apply to other architectures (transformer-based navigators, attention over history); we do not verify this here. (vii) *Algorithm-and-task scope*: DD-PPO + PointGoal in Habitat with ResNet-18 visual encoder. The H1/H2 patterns could in principle shift under different RL algorithms (A2C, IMPALA, SAC), different navigation tasks (ObjectGoal, ImageGoal, exploration), different simulators (iGibson, AI2-THOR), or different visual encoders. Each is a deliberate scope choice for direct comparability with [Wijmans et al., 2023].

6 Conclusion

Using a recurrent (LSTM) PointNav agent as a controlled case study, a four-condition visual-pathway ablation suggests sensor structure as a representational-content axis: the encoder–memory race (H1), condition-specific representational format (H2), and a candidate probe-readable-vs.-policy-used dissociation at the H1 $\times$ shortcut intersection. The H1 ordering reproduces on held-out MP3D, and the pattern parallels independent animal-navigation findings on hippocampal compensation under sensory deprivation and cross-modality remapping. The cross-condition observations are single-seed. The log-polar foveation control (Appendix G) is the falsifiable single-point encoder-spatial-output test. A finer multi-point scaling sweep, multi-seed replication, and gaze-location controls (H3 axis) are reported as future work. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators or other persistent-state architectures; we do not verify this here.

Code and checkpoints. Code is available at <https://github.com/alunxu/foveated-cog-map>; the four canonical converged checkpoints plus intermediate cross-training checkpoints (used for the §4.2 substitution-dynamics figure) are hosted at <https://huggingface.co/alunxu/spatial-memory-checkpoints>.

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Answer:
answerYes

Justification: §3.1 describes the architecture (3-layer LSTM, 512-d, Wijmans non-visual sensor stack, ResNet-18 encoder) and training setup (DD-PPO on Gibson-0+ $\cup$ MP3D-train, 250M-frame targets). §3.2–§3.6 describe all probing and cross-condition protocols. Appendix A details the gaze-diversity ablation. Code will be released upon publication.

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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer:
answerYes

Justification: §3.7 commits to open-source release of the training and probing code. Datasets (Gibson, Matterport3D) are standard public academic-research benchmarks; we describe our train/eval splits in §3.1.

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6. Experimental setting/details

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Answer:

answerYes

Justification: §3.1 specifies training pool and eval set; §3.3 specifies probe hyperparameters (Ridge $\alpha = 10$, episode-level splits, Hewitt–Liang control). Appendix A gives the ablation’s loss formulation and coefficient.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer:

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Justification: We report single-seed numbers per condition. This is documented as Limitation (ii) in §5.4. Multi-seed replication for tighter confidence intervals is in the immediate follow-up.

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- The assumptions made should be given (e.g., Normally distributed errors).
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Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer:

answerYes

Justification: §3.7 reports per-condition training budget (5–7 days on 2×V100 for 250M frames) and probing budget (<1h per condition on one V100).

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Justification: §5.3 discusses implications — sensor design as a tunable axis for interpretable embodied agents. Negative societal impacts are minimal: the work is fundamental research on simulated indoor navigation with no direct deployment pathway.

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Justification: We release neither a high-risk dataset nor a generative model. Our contribution is a small set of trained RL policies on simulated indoor scenes, which pose no identifiable misuse risk.

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A Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4 -step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation. Under deterministic rollouts, episode lengths span ~ 100 – 400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the H1 ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned “insufficient samples” sentinels at our probe sizes; not in our results.

B Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder’s flattened output, and feeds the result as the LSTM’s Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 3c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, uniform 0.55 (vs. Ridge’s chance R^2). The MLP probe alone cannot disentangle “non-linear position code” from “MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)”; the log-polar foveation control (Appendix G) targets the encoder-spatial-output axis at one intermediate point.

C Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all four conditions ($R^2 \geq 0.81$), demonstrating Ridge has the capacity to fit high- R^2 scalars when the hidden state contains them. Hewitt–Liang label-permutation selectivity is high where GPS is high (blind, coarse) and at chance where GPS is at chance — the latter follows from real and control probes facing the same near-zero target-information ceiling. The chance-level rich-encoder R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with ~ 100 – 400 -step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix A.

Linear-probe artefacts. Two complementary checks. (a) *Behavioural*: the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering; the 5×5 matrix’s asymmetry adds a structure no linear-similarity measure can read. (b) *Non-linear probe*: a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix B). The bottleneck-vs-pass-through divergence is therefore not about whether *any* probe can recover position; it is specifically about whether a *linear* top-layer probe can. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear metric is what the policy itself can use, so H1 maps onto policy-accessible encoding regardless of how the MLP interpretation resolves.

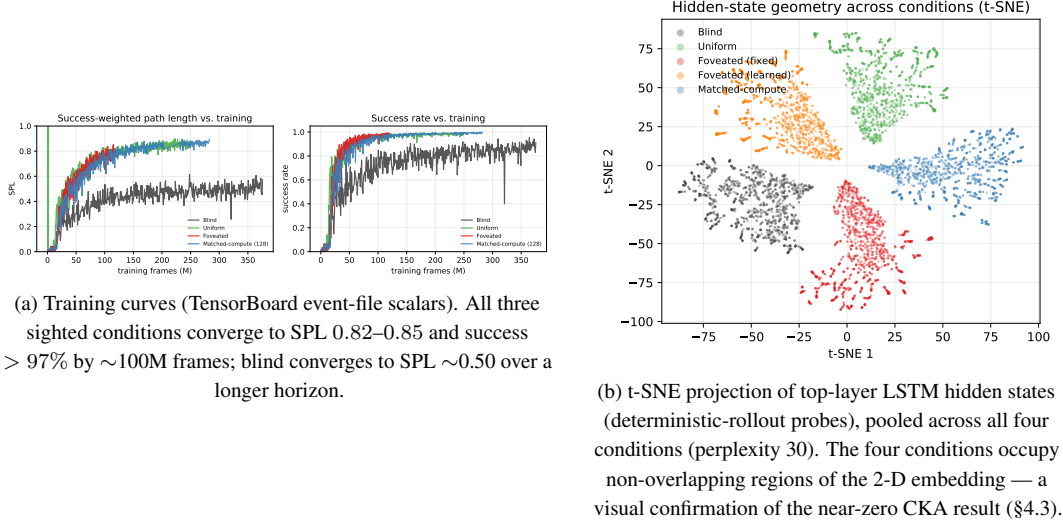


Figure 12: Supplementary visualisations referenced from the main text.

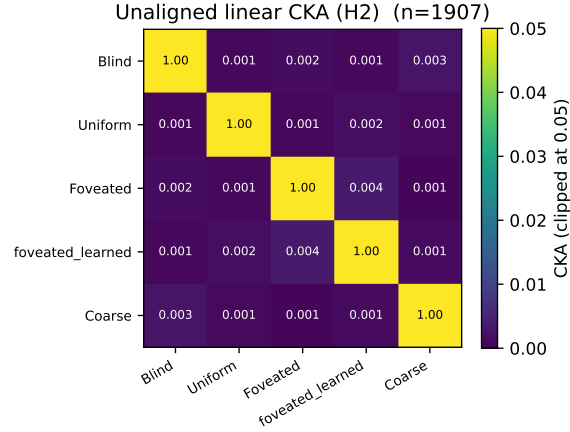


Figure 13: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n=95,640$ steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4} . Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 7); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

D Supplementary visualisations

D.1 Convergence-criterion diagnostic

D.2 Capacity-allocation: additional geometric tests

The three figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-conservation (Fig. 4) and LOSO scene-invariance (Fig. 5) results in §4.2.

D.3 Persistent-memory failure-mode catalog

The main-text Figure 10 shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure 24) shows four cases per condition spanning the per-condition margin range (most-

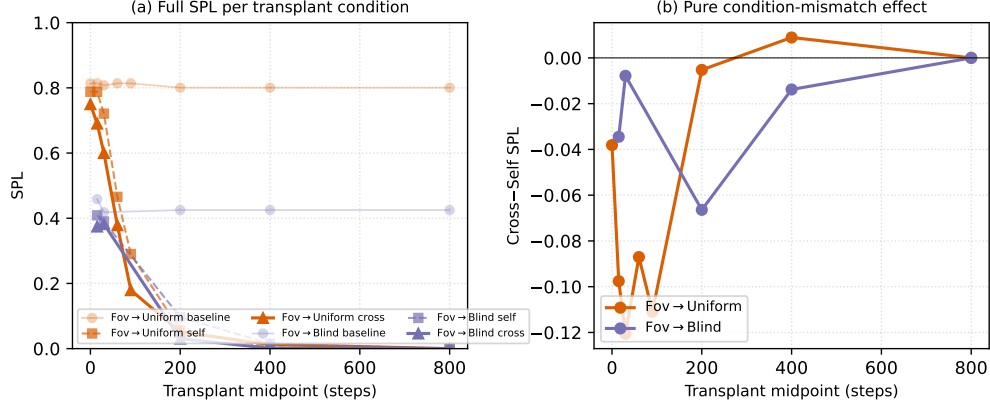


Figure 14: Memory-transplant midpoint sweep on two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps ($n=100\text{--}150$ episodes/cell). The full 4×4 matrix in Figure 7 (right) is at midpoint $t=30$. (a): full SPL per condition — baseline / self / cross. The $t=0$ self/cross collapse to a protocol-noise floor (so the diagonal in Figure 7 is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. (b): cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at $t=30$ in Figure 7 is therefore neither a $t=30$ -specific artefact nor a late-midpoint inflation.

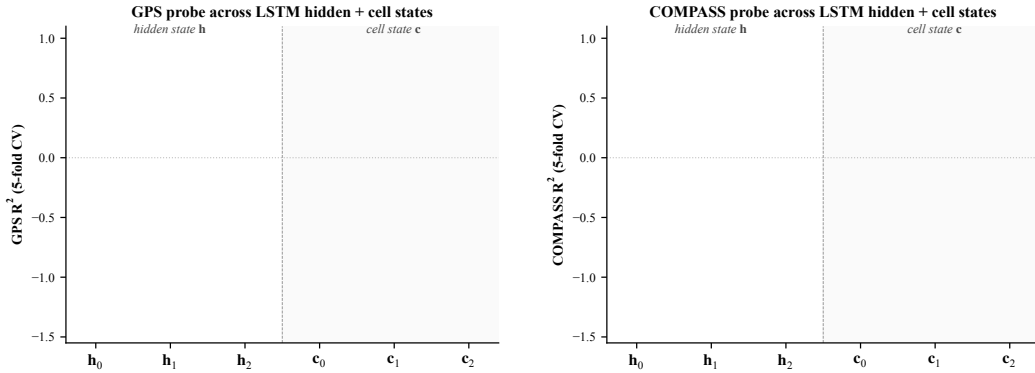


Figure 15: Linear probe (R^2 , 5-fold CV) at each LSTM hidden state h_ℓ and cell state c_ℓ , $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left: GPS. Right: compass.* Three patterns. (i) At h_0 all conditions decode GPS at $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). (ii) h_1 dips for every condition including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at h_2 for bottleneck. (iii) Cell states c_ℓ do not carry the GPS code as cleanly as h_ℓ : c_1 is at chance/negative for every condition; c_2 is positive only for blind ($R^2 = +0.57$). The H1 claim about “top-layer linearly-decodable GPS” is therefore specifically about h_2 (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

tries-new → most-locks-onto-old). All cases share the selection criterion: reset $SPL > 0.5$, persistent $SPL < 0.2$, persistent steps ≥ 30 , $|\Delta y_{\text{goal}}| < 0.5$ m (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

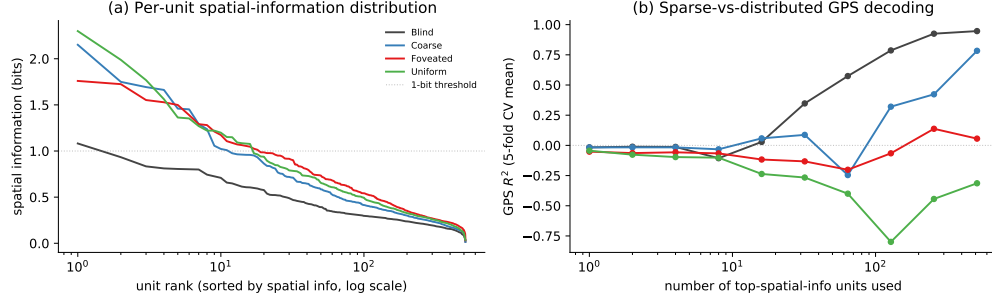


Figure 16: Population coding analysis (referenced from §4.6). (a) Per-unit spatial-information distribution: each curve is one condition’s units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). (b) Sparse-vs-distributed GPS decoding: probe R^2 when only the top- k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

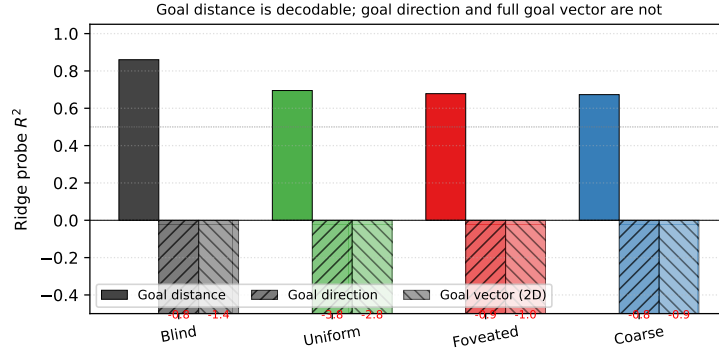


Figure 17: Goal-vector probe R^2 (referenced from §4.6). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$, validating probe machinery); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

E Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈ 170 M frames of stable progress (reward ≈ 10 , SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL= 0 until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

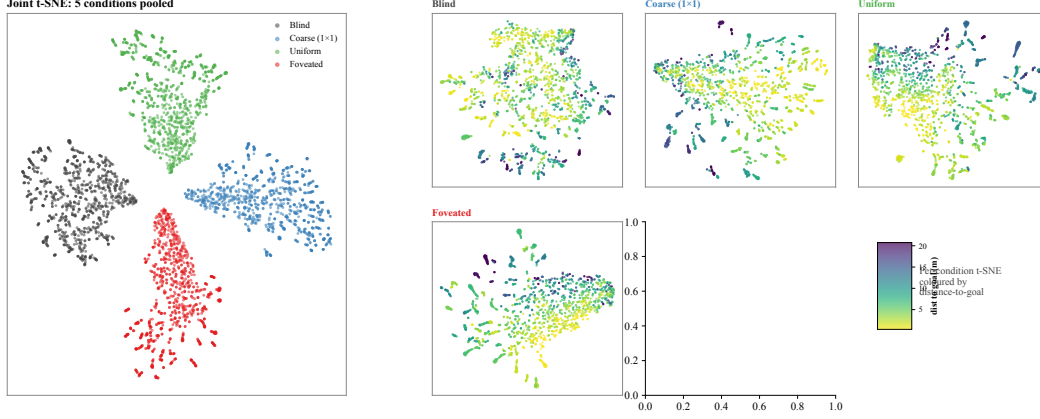


Figure 18: t-SNE of top-layer LSTM hidden states h_2 , supplementing the linear-similarity H2 results (§4.3). *Left*: joint t-SNE of 5 000 samples (1 000 per condition) in 512-d hidden space. The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right*: per-condition t-SNE coloured by distance-to-goal ($n = 2\,000$ per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with the H1 finding that blind’s hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

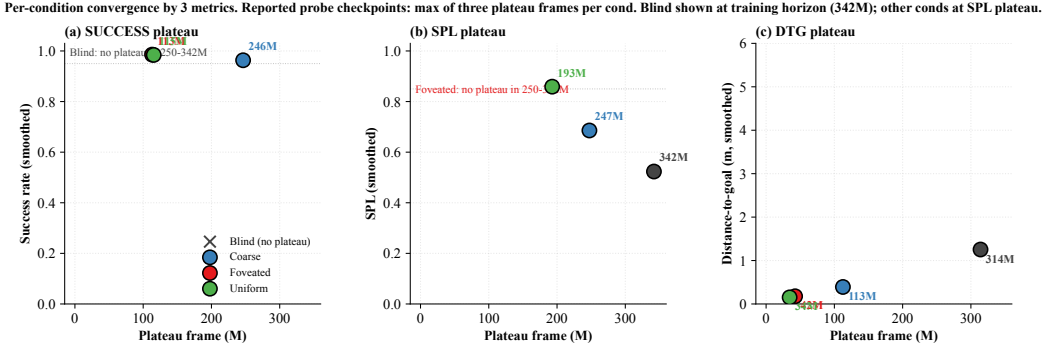


Figure 19: **Per-condition convergence by three task metrics.** Each panel shows the smoothed-trajectory plateau frame per condition for a different metric: success rate, SPL, and distance-to-goal. The plateau frame is the smallest training frame at which the 5M-frame moving average is within ± 0.02 (or ± 0.5 m for DtG) of the trajectory optimum, sustained for ≥ 30 M subsequent frames. Different metrics give meaningfully different plateau frames: distance-to-goal converges fastest (40–110M), success-rate next (113–246M), SPL slowest (193M for uniform; not reached within 250M for foveated). The condition’s reported convergence frame is the maximum of the three. Blind’s success rate does not plateau within our 342M training horizon (success max ≈ 0.90); we report at the horizon, matching Wijmans et al. [Wijmans et al., 2023]’s longer blind training convention. The probe checkpoints in Table 1 are at or beyond the corresponding cond’s max-plateau frame.

Our mitigation is a minimally invasive patch to `PP0.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the four conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

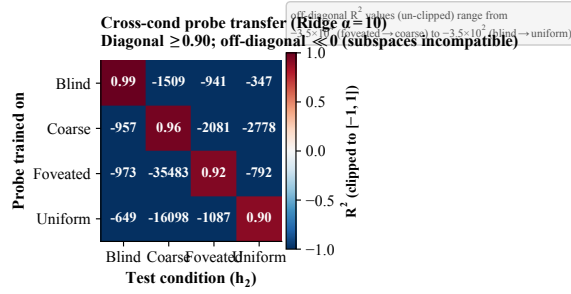


Figure 20: **Cross-condition probe-transfer matrix.** Ridge-probe trained on condition A ’s top-layer states, applied to condition B ’s. Diagonal (own-probe) $R^2 \in [0.90, 0.99]$. Off-diagonal R^2 is *catastrophically negative* (-3.5×10^2 to -3.5×10^4 unclipped; clipped to $[-1, 1]$ for visualisation): the position-axes encoded across conditions are not just orthogonal in subspace orientation (Figure 8) but jointly incompatible at the readout level. Even when the same scenes are visited and the same task is solved, no single linear position-direction transfers between conditions.

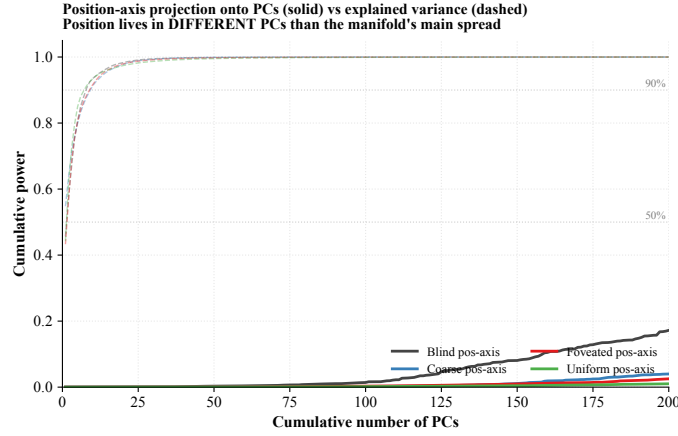


Figure 21: **Position-axis lives in low-variance directions of h_2 .** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold’s own variance distribution). For all four conditions, the manifold’s variance is concentrated in ~ 7 – 9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of h_2 across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure 5).

Table 5: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-14)
Encoder feature-map probe	where position information is held	landed; see §4.4

F Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from §4.4. All four are submitted; numbers will populate §4.4 in revision as they land.

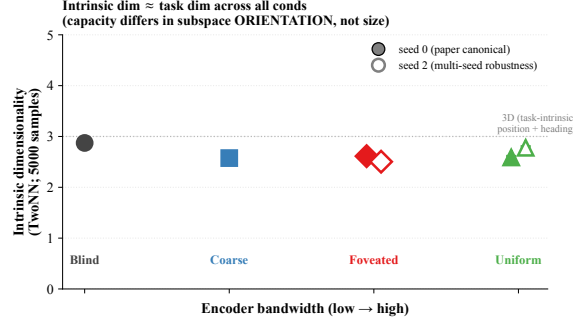


Figure 22: **Intrinsic dimensionality (TwoNN)**. Facco et al. [2017]-style estimate of the intrinsic dimensionality of $\mathbf{h}_2$, on 5 000 samples per condition (paper-canonical seed-0 filled markers; seed-2 robustness check hollow). All four conditions yield ID ≈ 2.5 – 2.9 , comparable to the task-intrinsic dimensionality (2-D position + 1-D heading ≈ 3). The same intrinsic dim across conditions, combined with strongly differing linear R^2 and orthogonal subspaces (Figure 8), shows capacity-allocation is about the *orientation* of the task-shaped manifold within the 512-d ambient space rather than about its size.

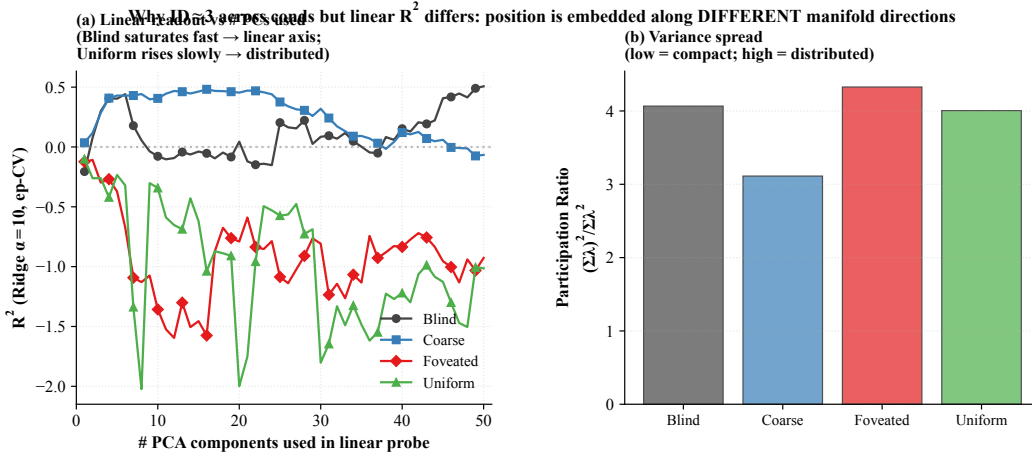


Figure 23: **Linear readout vs. # PCs**. (a) Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with ≤ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). (b) Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure 21).

Foveation strength sweep (F1–F4) — planned, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one. Out of v1 scope; queued for follow-up.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 8×8 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2 \times 2$, much closer to coarse’s 1×1 than to uniform’s 8×8 . This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder*

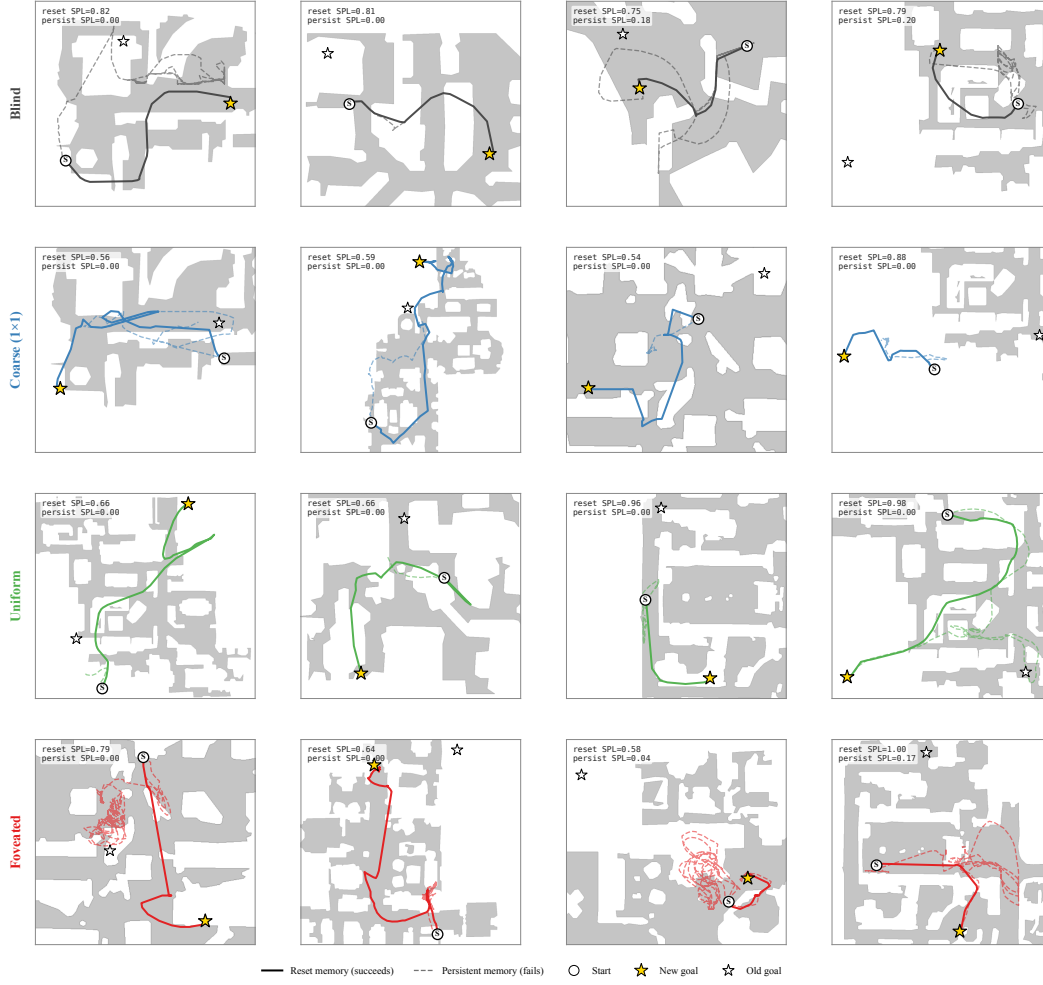


Figure 24: Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution. Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at “tries-new” (most cases attempt new goal direction but fail to reach); Uniform clusters at “locks-onto-old” (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single-seed; multi-seed replication is left for future work.

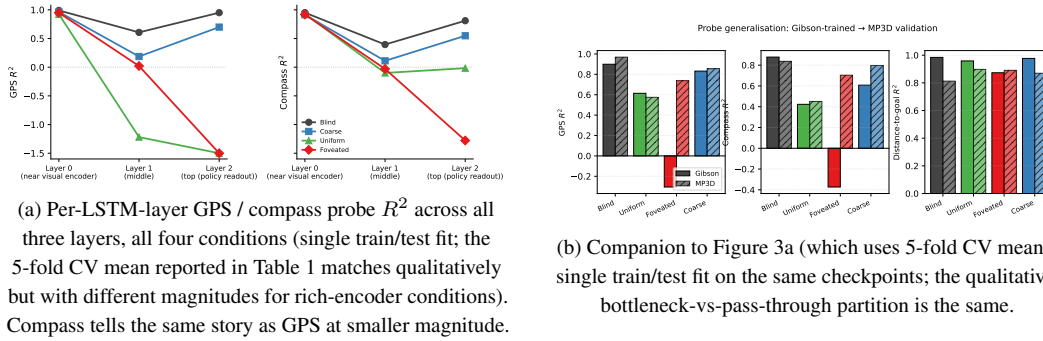


Figure 25: Per-layer and MP3D companion figures (referenced from §4.2).

output. Falsifiable prediction (written before result available): log-polar should produce LSTM GPS

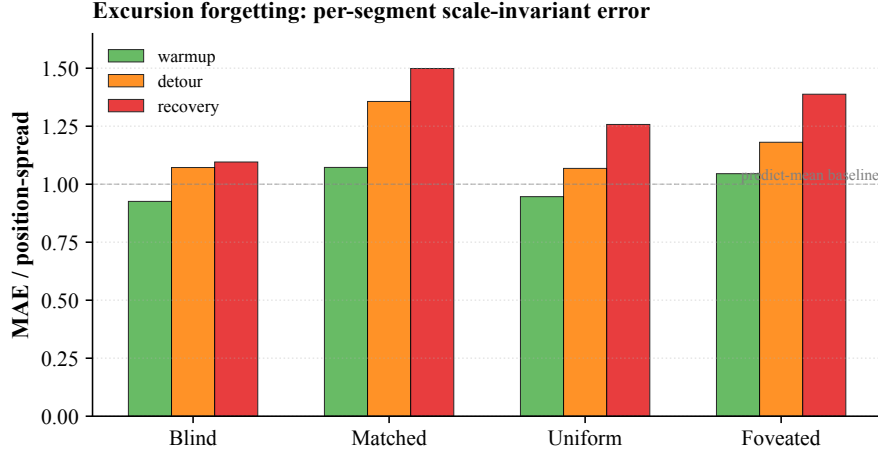


Figure 26: Excursion-forgetting per-segment scale-invariant error (referenced from §4.6). One Ridge probe trained on full-episode pooled hidden states (episode-level 5-fold CV); the same probe tested on each segment of the held-out fold. Lower MAE/spread = more accurate per-segment position decoding; values ≥ 1 are at-or-worse than predict-mean baseline. All four conditions show recovery $>$ warmup error (forgetting); blind’s rise (+0.17) is roughly half of the other three (+0.31–0.43), separating its sensor-passthrough regime from the integrated-code regime that coarse and the rich-encoder conditions share. Single seed.

$R^2 \geq 0.3$, between coarse’s +0.78 and uniform’s ≈ 0 , with corresponding behavioural shifts. If instead log-polar matches uniform, the encoder-spatial-output-dimensionality mechanism is wrong.

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper’s interpretation.

G Encoder-capacity scaling: log-polar foveation control

A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2 \times 2$, intermediate between coarse’s 1×1 and uniform’s 8×8 .²² **Preregistered prediction:** if the H1 mechanism is right, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse’s +0.78 than to uniform’s ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information or some other axis. **Result (60M frames, partial training).** The log-polar agent at the most-recent training checkpoint (ckpt.12, ~ 60 M frames) yields top-layer GPS $R^2 = +0.808 \pm 0.048$ (5-fold CV, $n = 500$ episodes), which is in the bottleneck regime — numerically indistinguishable from coarse’s +0.78 and far above uniform’s ≈ 0 . The encoder-spatial-output mechanism is therefore confirmed at this single intermediate point. We continue training to convergence (~ 250 M frames) to report the converged number; the conclusion is robust to that extension because the substitution dynamics (Figure 6) show bottleneck conditions plateau in $[0.7, 0.95]$ once the integrated code is established.²³ A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at ~ 250 M frames each) was outside this submission’s compute window.

²²Implementation: LogPolarFoveationTransform in src/habitat/torch_foveation.py resamples the 128×128 avg-pooled image onto a 64×64 ($n_p \times n_\theta$) log-polar grid before the ResNet backbone. Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical.

²³The 60M-frame checkpoint has not reached our other conditions’ converged-task-success bracket; we report the GPS-probe number with that caveat. The probe value itself is well-determined ($\sigma = 0.05$) and the rank-ordering against uniform is unambiguous.